Particles and Nuclei 2

- 3/31/2026
- Announcements:

- Shorter HW 5 released due date

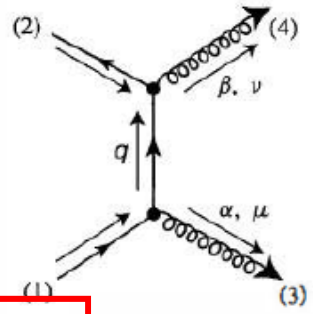
- Reading – Just continue Mann Ch 7 and 8

(Detectors Mann Chapter 8)

Gluon Color factor cases→COMPTON

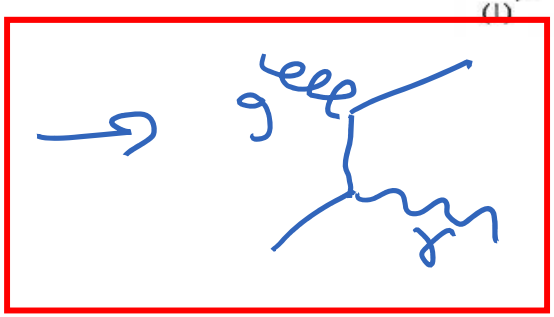
- qqbar

$$\mathcal{M}_1 = \frac{-g_s^2}{8}\frac{1}{p_1 \cdot p_3}\left\{\bar{u}(2)[\not\epsilon_4(\not p_1 - \not p_3 + mc)\not\epsilon_3]u(1)\right\} a_3^\alpha a_4^\beta (c_2^\dagger \lambda^\beta \lambda^\alpha c_1)$$



- Gluon Compton
  - Start small





10.4 The Gluon Emission Cross Section

We are now ready to include the gluon emission diagrams of Fig. 10.2. To calculate $\gamma^* q \to qg$, we take over our results for Compton scattering. In Section 6.14, we evaluated the closely related QED process $\gamma^* e \to \gamma e$ and found

$$\overline{|\mathcal{M}|^2} = 32\pi^2\alpha^2\left(-\frac{u}{s} - \frac{s}{u} + \frac{2tQ^2}{su}\right), \qquad (10.16)$$

Cross Sections → Some Qualitatives

2 *50. Cross-section formulae for specific processes*

PDG:

$$\frac{d\sigma}{d\Omega}(q\overline{q} \to q'\overline{q}') = \frac{\alpha_s^2}{9s}\frac{t^2+u^2}{s^2} \ . \tag{50.4}$$

Crossing symmetry gives

$$\frac{d\sigma}{d\Omega}(qq' \to qq') = \frac{\alpha_s^2}{9s}\frac{s^2+u^2}{t^2} \ . \tag{50.5}$$

If the quarks q and q' are identical, we have

$$\frac{d\sigma}{d\Omega}(q\overline{q} \to q\overline{q}) = \frac{\alpha_s^2}{9s}\left[\frac{t^2+u^2}{s^2} + \frac{s^2+u^2}{t^2} - \frac{2u^2}{3st}\right] \ , \tag{50.6}$$

and by crossing

$$\frac{d\sigma}{d\Omega}(qq \to qq) = \frac{\alpha_s^2}{9s}\left[\frac{t^2+s^2}{u^2} + \frac{s^2+u^2}{t^2} - \frac{2s^2}{3ut}\right] . \tag{50.7}$$

Annihilation of e^+e^- into $\gamma\gamma$ has the cross section

$$\frac{d\sigma}{d\Omega}(e^+e^- \to \gamma\gamma) = \frac{\alpha^2}{2s}\frac{u^2+t^2}{tu} \ . \tag{50.8}$$

The related QCD process also has a triple-gluon coupling. The cross section is

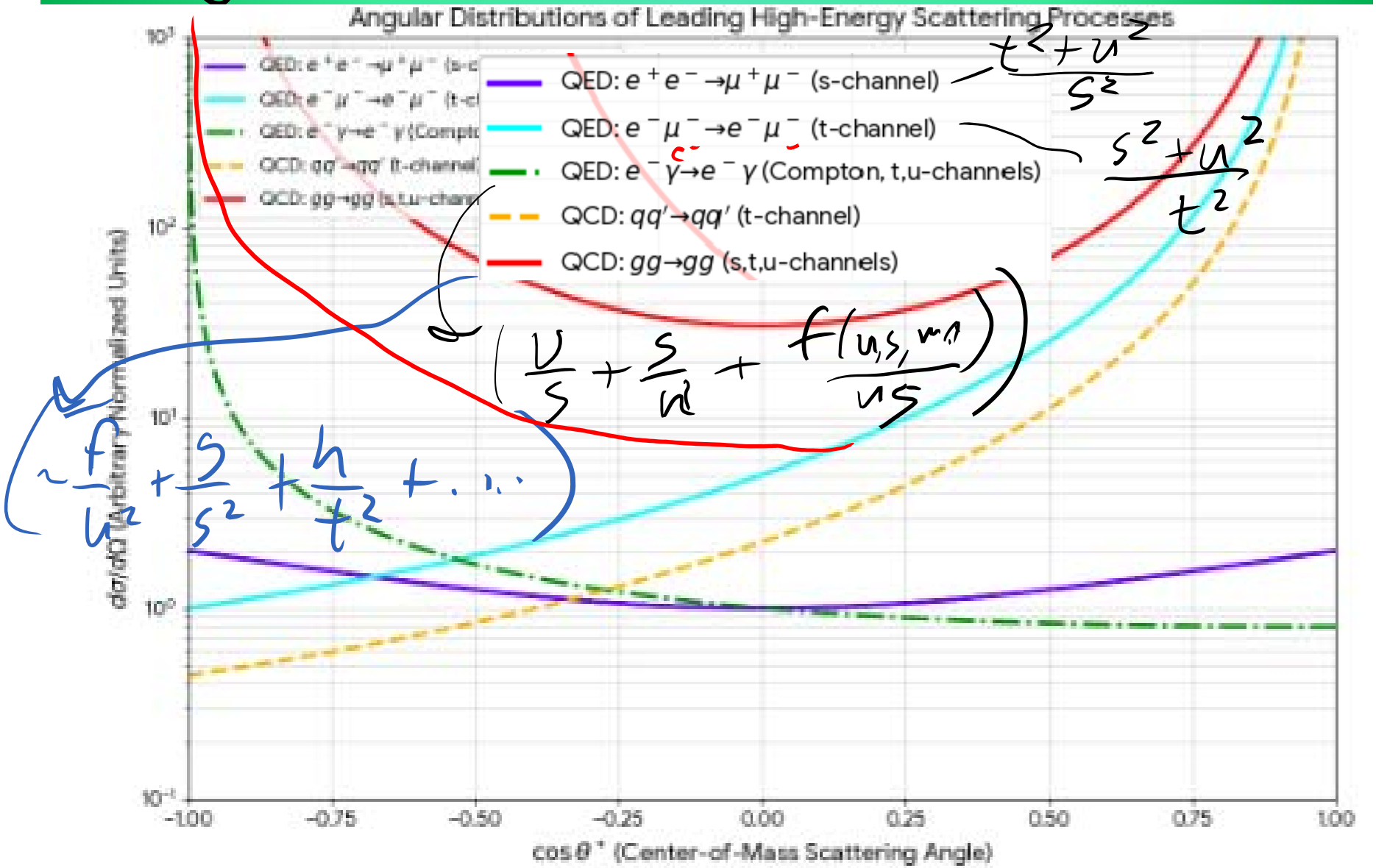
$$\frac{d\sigma}{d\Omega}(q\overline{q} \to gg) = \frac{8\alpha_s^2}{27s}(t^2+u^2)\left(\frac{1}{tu} - \frac{9}{4s^2}\right) \ . \tag{50.9}$$

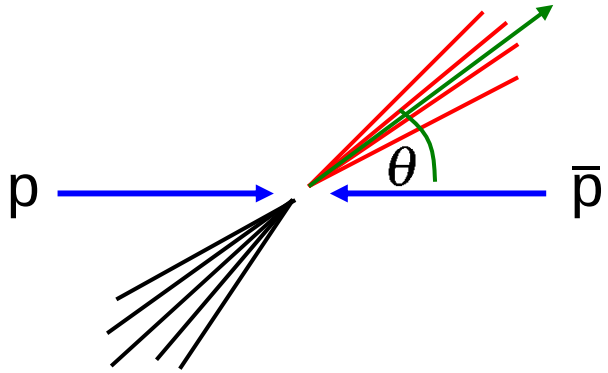
The crossed reactions are

$$\frac{d\sigma}{d\Omega}(qg \to qg) = \frac{\alpha_s^2}{9s}(s^2+u^2)(-\frac{1}{su} + \frac{9}{4t^2}) \tag{50.10}$$

- We've learned how to calculate many cross section results from fundamental QED and QCD
- What do the formula imply physically.
- Angular dependences is one way to get some insight

Angular Distributions from Mandel's

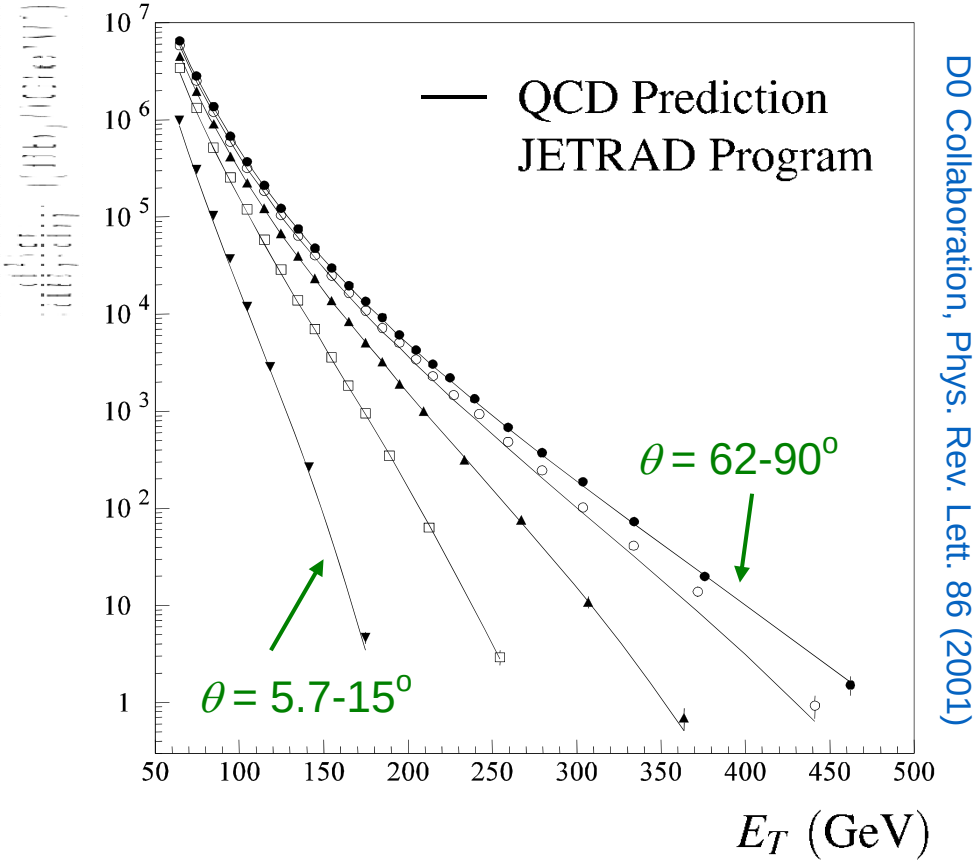




★ Measure cross-section in terms of

- **"transverse energy"** $E_T = E_{\text{jet}} \sin\theta$
- **"pseudorapidity"** $\eta = \ln\left[\cot\left(\frac{\theta}{2}\right)\right]$

…don't worry too much about the details here, what matters is that…



D0 Collaboration, Phys. Rev. Lett. 86 (2001)

★QCD predictions provide an excellent description of the data

★NOTE:

- at low E_T cross-section is dominated by low *x* partons

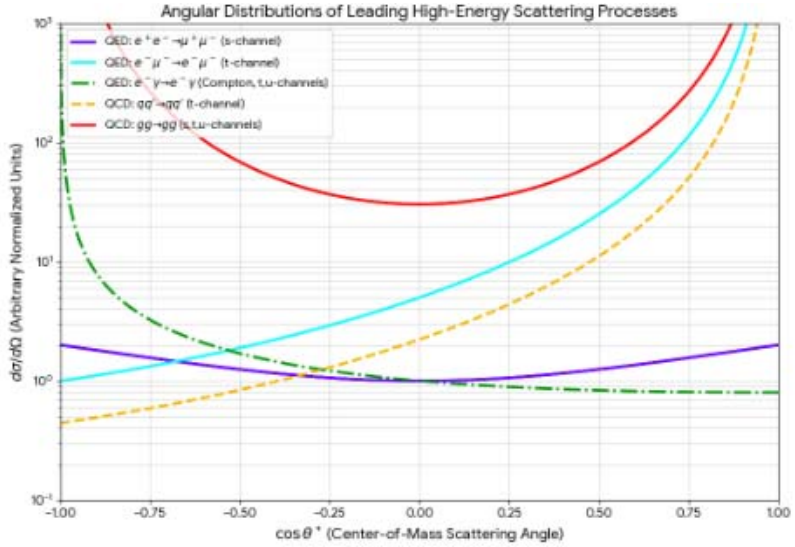
  i.e. gluon-gluon scattering

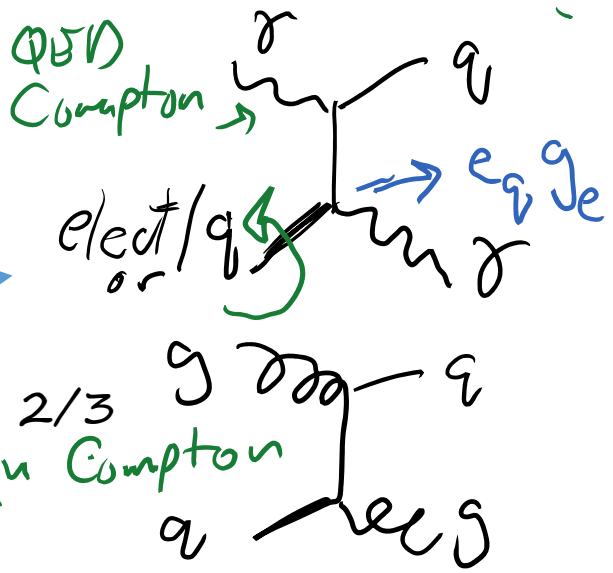
- at high E_T cross-section is dominated by high *x* partons

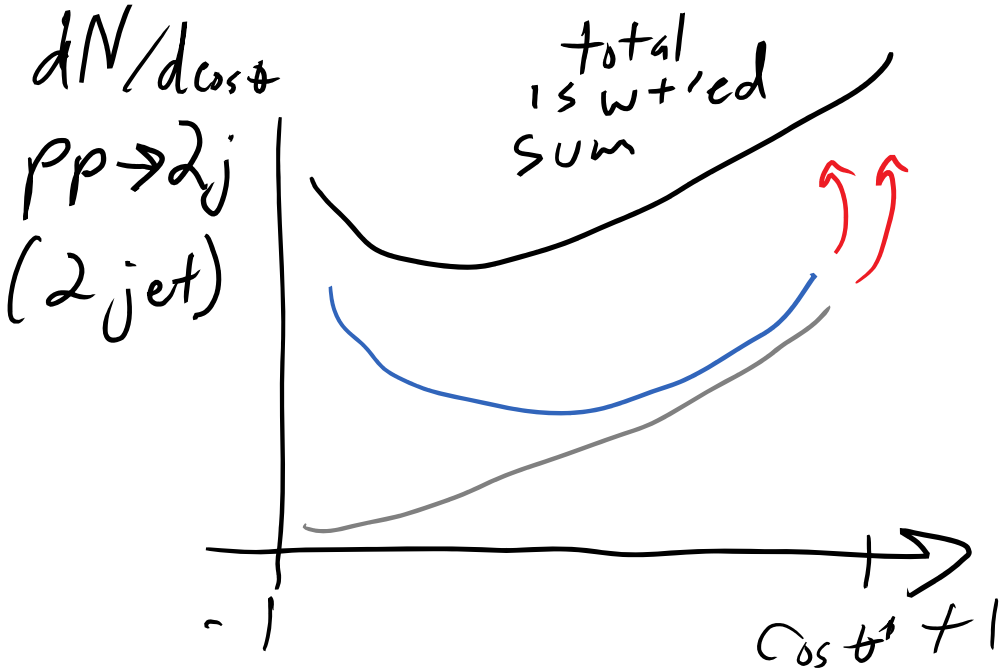
  i.e. quark-antiquark scattering

USING these calcs: Wt'd Sums

- So Example Observable say for 2 –jet events
  - p+p → 2j → parton+parton→ 2j
- Must put together all these processes w/ weightings
- Coupling's α's :
  - QED processes get α_{EM} & sometimes factors of fractional ch eq (vertex factor $g_e \rightarrow e_q g_e$ e.g. $e_q \propto 2/3$
  - QCD αstrong
- CF : Within QCD processes
  - COLOR FACTOR DETERMINES

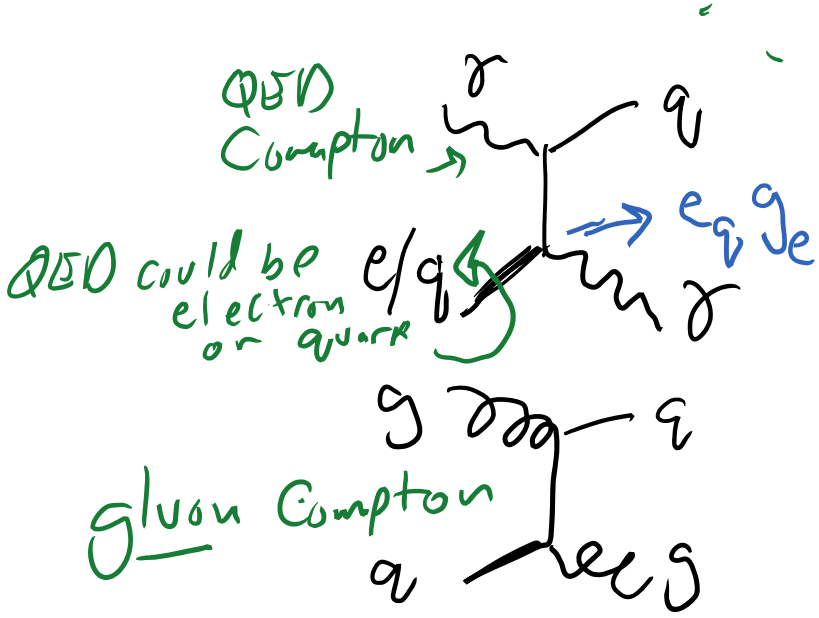


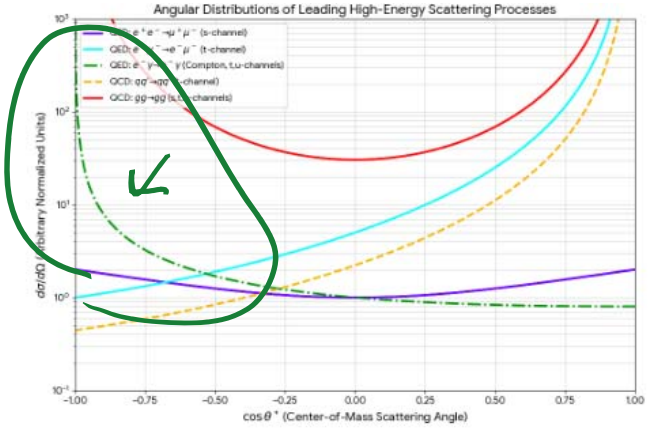




Compton Process Important

- Compton Edge in Low E
- BASIC DIAGRAM FOR BREMSTRAHLUNG
  - Understand "Negative angular dependence electron CONVERTS to PHOTON (photon in same direction as initial e's mom, scattered e goes in NEGATIVE DIRECTION wrt to initial electron
- Jet production: most probability in pdf's is for GLUON pdfs at low x → low fraction of proton momentum (see later lecture

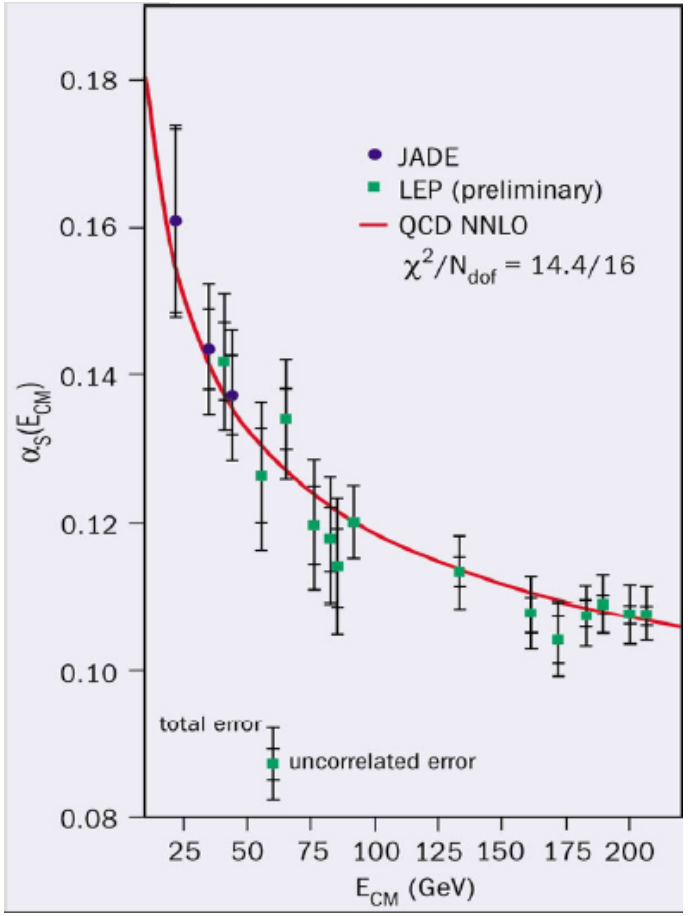




See slides last time

Note

- Alphas(Q^2)
- Depending on the Order (NLO) of the Calc some fraction of Alpha_s running is Encoded in PDF's



One MORE thing from Ch 7 and Ch 8 qcd

RENORMALIZATION

Renormalization "Group" equation
Renormalization SCHEME : MS Scheme

Griffiths Chapter 7 Qed

.....Loop diagrams, renorm math tricks:

"Running Coupling: → "SOP UP INFINITY IN RESCALED CHARGE/COUPLING → induces Q^2 dependence on alpha

$$\alpha(|q^2|) = \alpha(0)\left\{1 + \frac{\alpha(0)}{3\pi}\ln(|q^2|/(mc)^2)\right\} \quad (|q^2| = -q^2 \gg (mc)^2)$$

QED coupling increases with q^2

Exercise

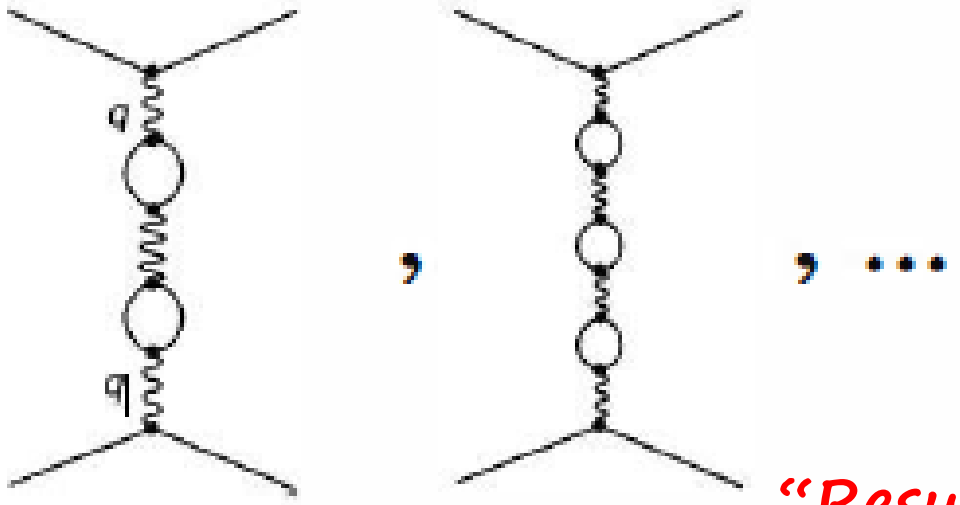
- Calc alpha_s at 1 GeV, 10 GeV, 100 GeV how large is NLO compared to LO

- Use Λ – 0.3 gev try 1 w/ using it as 1 GeV

$$\alpha_s(|q^2|) = \frac{12\pi}{(11n - 2f)\ln(|q^2|/\Lambda^2)} \qquad (|q^2| \gg \Lambda^2)$$

Renorm in QCD

- Asymptoptic Freedom -- alpha smaller w/ q^2



"Resummation"

As it happens, these can be summed explicitly, and the result is†

$$\alpha(|q^2|) = \frac{\alpha(0)}{1 - [\alpha(0)/3\pi]\ln[|q^2|/(mc)^2]} \qquad (|q^2| \gg (mc)^2) \qquad (8.92)$$

"Factorizing" Photon polariz/color

$$\mathcal{M}_2 = \frac{g_e^2}{(p_1+p_2)^2 - m^2c^2}\left[\bar{u}(4)\not{\varepsilon}(3)^*(\not{p}_1 + \not{p}_2 + mc)\not{\varepsilon}(2)u(1)\right]$$

- Sum over all spin states of photon
- after all the trace theorems it looks like this because we can factor out both polarization vectors

$$\sum_{\gamma\,spin} \varepsilon^*(3)_\mu \varepsilon(2)_\nu \, T^{\mu\nu}$$

Completeness for photons two times

$$\sum_{\gamma\,spin}' \varepsilon^*(3)_\mu \varepsilon(2)_\nu \sum_{\gamma'\,spin} \varepsilon_\kappa(3)\varepsilon^*_{\alpha'}(2)\, T^{\mu\nu\kappa\alpha'}$$

"$Q_{\mu\kappa} Q_{\nu\alpha}$"

$$T = \mathrm{Tr}\left[\gamma^\mu(\not{p}_1+\not{p}_2)\gamma^\nu \not{p}_1 \gamma^\kappa(\not{p}_1+\not{p}_2)\gamma^\alpha \not{p}_4\right]$$

This is the answer to the M_2 part of Griff Prob. 7.32 – also shown in slides is H&M version–compare, I said they look like they're ~the same (to a~sign)

Color Factors

Most simple QCD diagrams are exactly analogous to QED diagrams, and the QCD cross section is obtained by the replacement

$$\alpha^n \to C_F \alpha_s^n, \tag{10.18}$$

Quark and Quark

We turn now to the interaction of two quarks. Again, we shall assume that they have different flavors, so the only diagram (in lowest order) is the one indicated in Figure (8.7),* representing, say, $u + d \to u + d$. The amplitude is

$$\mathcal{M} = \frac{-g_s^2}{4}\frac{1}{q^2}[\bar{u}(3)\gamma^\mu u(1)][\bar{u}(4)\gamma_\mu u(2)](c_3^\dagger \lambda^\alpha c_1)(c_4^\dagger \lambda^\alpha c_2) \tag{8.56}$$

Color Factors

- For simple cases (gluon propagators/quark)

3. *Vertices.* Each vertex introduces a factor

Qua k-gluon:  : $\frac{-ig_s}{2}\lambda^\alpha\gamma^\mu$ (8.42)

Origin of this extra factor of ½ compared to QED ? PURE CONVENTION! [implication comparing α_s to α]

$$\mathscr{M} = -\frac{g_s^2}{4Q^2}[\bar{u}(3)\gamma^\mu u(1)][\bar{v}(2)\gamma_\mu v(4)](c_3^\dagger\lambda^\alpha c_1)(c_2^\dagger\lambda^\alpha c_4)$$

$$C_F = \frac{1}{4}(\quad)$$

So the CF's (tree level above) always have factor of ¼ to start with

QCD Color objects

- various

$$c = \begin{pmatrix}1\\0\\0\end{pmatrix} \text{ for red, } \begin{pmatrix}0\\1\\0\end{pmatrix} \text{ for blue, } \begin{pmatrix}0\\0\\1\end{pmatrix} \text{ for green}$$

NOTICE "RBG" NOT normal RGB

$$\lambda^1 = \begin{pmatrix}0 & 1 & 0\\1 & 0 & 0\\0 & 0 & 0\end{pmatrix} \quad \lambda^2 = \begin{pmatrix}0 & -i & 0\\i & 0 & 0\\0 & 0 & 0\end{pmatrix} \quad \lambda^3 = \begin{pmatrix}1 & 0 & 0\\0 & -1 & 0\\0 & 0 & 0\end{pmatrix}$$

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Particles and Nuclei 2

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- Reading – TBD will email tonight

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$$\lambda^7 = \begin{pmatrix}0 & 0 & 0\\0 & 0 & -i\\0 & i & 0\end{pmatrix} \quad \lambda^8 = \frac{1}{\sqrt{3}}\begin{pmatrix}1 & 0 & 0\\0 & 1 & 0\\0 & 0 & -2\end{pmatrix}$$

Simple Cases

- Octet states for q qbar (same as gluon octet)

$$\left\{\begin{array}{ll} |1\rangle = (r\bar{b} + b\bar{r})/\sqrt{2} & |5\rangle = -i(r\bar{g} - g\bar{r})/\sqrt{2} \\ |2\rangle = -i(r\bar{b} - b\bar{r})/\sqrt{2} & |6\rangle = (b\bar{g} + g\bar{b})/\sqrt{2} \\ |3\rangle = (r\bar{r} - b\bar{b})/\sqrt{2} & |7\rangle = -i(b\bar{g} - g\bar{b})/\sqrt{2} \\ |4\rangle = (r\bar{g} + g\bar{r})/\sqrt{2} & |8\rangle = (r\bar{r} + b\bar{b} - 2g\bar{g})/\sqrt{6} \end{array}\right\} \qquad (8.29)$$

- Singlet state (actual mesons – no such gluon)

$$|9\rangle = (r\bar{r} + b\bar{b} + g\bar{g})/\sqrt{3}$$

- qq (e.g. 2 of baryon 3)

$$\left\{\begin{array}{l} (rb - br)/\sqrt{2} \\ (bg - gb)/\sqrt{2} \\ (gr - rg)/\sqrt{2} \end{array}\right\} \quad \text{(triplet)} \qquad \left\{\begin{array}{l} rr, bb, gg, \\ (rb + br)/\sqrt{2}, (bg + gb)/\sqrt{2}, (gr + rg)/\sqrt{2} \end{array}\right\} \quad \text{(sextet)}$$

Griffiths Example 8.1

- qqbar octet example pick incoming p1- R p2Bbar

$$C_F = \frac{1}{4}\left(R\,\lambda^{\alpha}\,\bar{R}\right)\left(B\,\lambda^{\alpha}\,\bar{B}\right)$$

$$\frac{1}{4}\,\lambda^{\alpha}_{ij}\,\lambda^{\alpha}_{kl}$$

- Will equal
- Calculate this color factor remember you must sum over all gluons

Griffiths Example 8.1

- qqbar octet example pick incoming p1- R p2Bbar

$$C_F = \frac{1}{4}(R\lambda^\alpha \bar{R})(B\lambda^\alpha \bar{B})$$

$$\frac{1}{4}\lambda^\alpha_{ij}\lambda^\alpha_{kl}$$

- Will equal
- Calculate this color factor

$$f = \frac{1}{4}\left[(1\,0\,0)\lambda^\alpha\begin{pmatrix}1\\0\\0\end{pmatrix}\right]\left[(0\,1\,0)\lambda^\alpha\begin{pmatrix}0\\1\\0\end{pmatrix}\right] = \frac{1}{4}\lambda^\alpha_{11}\lambda^\alpha_{22}$$

Sum over

Exercise

- tophat :which matrices will have non-zero values for these indice combos? answer 1,2,3....

$$\lambda^1 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad \lambda^2 = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad \lambda^3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\lambda^4 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} \quad \lambda^5 = \begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{pmatrix} \quad \lambda^6 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$\lambda^7 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix} \quad \lambda^8 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix}$$

8.19 Color factors always involve expressions of the form $\lambda^{\alpha}_{ij}\lambda^{\alpha}_{kl}$ (summed over α). There is a simple formula for this quantity, which shortens the arithmetic:

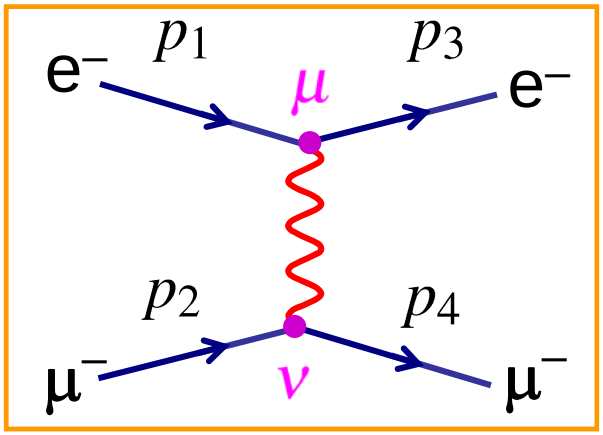
$$\lambda^{\alpha}_{ij}\lambda^{\alpha}_{kl} = 2\delta_{il}\delta_{jk} - \frac{2}{3}\delta_{ij}\delta_{kl}$$

QCD vs QED

QED

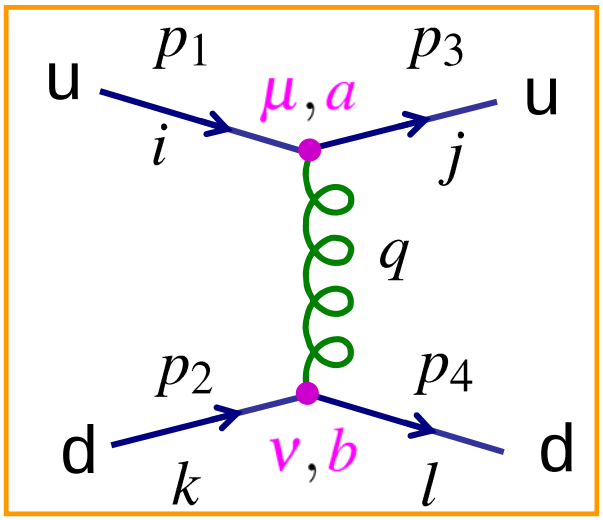
$$-iM = [\bar{u}(p_3) ie\gamma^\mu u(p_1)] \frac{-ig_{\mu\nu}}{q^2} [\bar{u}(p_4) ie\gamma^\nu u(p_2)]$$

$$M = -e^2 \frac{1}{q^2} g_{\mu\nu} [\bar{u}(p_3)\gamma^\mu u(p_1)][\bar{u}(p_4)\gamma^\nu u(p_2)]$$



QCD

$$M = -\frac{g_s^2}{4} \lambda^a_{ji} \lambda^a_{lk} \frac{1}{q^2} g_{\mu\nu} [\bar{u}_u(p_3)\gamma^\mu u_u(p_1)][\bar{u}_d(p_4)\gamma^\nu u_d(p_2)]$$



★ **QCD Matrix Element = QED Matrix Element with:**

- $e^2 \to g_s^2$ **or equivalently** $\alpha = \frac{e^2}{4\pi} \to \alpha_s = \frac{g_s^2}{4\pi}$

+ QCD Matrix Element includes an additional "colour factor"

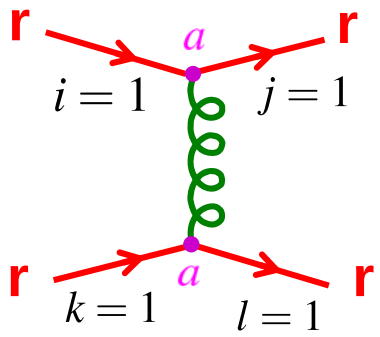
$$C(ik \to jl) \equiv \frac{1}{4} \sum_{a=1}^{8} \lambda^a_{ji} \lambda^a_{lk}$$

Evaluation of QCD Colour Factors

- **QCD colour factors reflect the gluon states that are involved**

$\lambda^1 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$	$\lambda^4 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}$	$\lambda^6 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$	$\lambda^3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$
$\lambda^2 = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$	$\lambda^5 = \begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{pmatrix}$	$\lambda^7 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}$	$\lambda^8 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix}$
Gluons: $r\bar{g}, g\bar{r}$	$r\bar{b}, b\bar{r}$	$g\bar{b}, b\bar{g}$	$\frac{1}{\sqrt{2}}(r\bar{r} - g\bar{g})$ $\frac{1}{\sqrt{6}}(r\bar{r} + g\bar{g} - 2b\bar{b})$

❶ Configurations involving a single colour



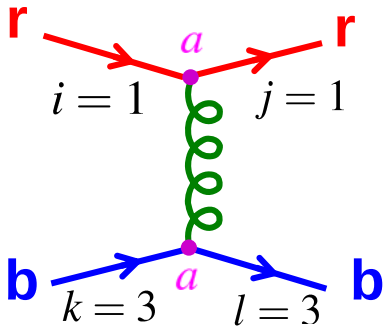
- **Only matrices with non-zero entries in 11 position are involved**

$$C(rr \to rr) = \frac{1}{4}\sum_{a=1}^{8} \lambda^a_{11}\lambda^a_{11} = \frac{1}{4}(\lambda^3_{11}\lambda^3_{11} + \lambda^8_{11}\lambda^8_{11})$$

$$= \frac{1}{4}\left(1 + \frac{1}{3}\right) = \frac{1}{3}$$

Similarly find $C(rr \to rr) = C(gg \to gg) = C(bb \to bb) = \frac{1}{3}$

❷ Other configurations where quarks don't change colour e.g. $rb \to rb$



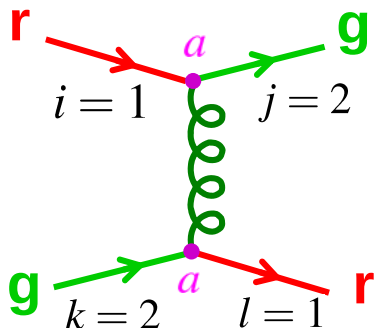
- Only matrices with non-zero entries in **11** and **33** position are involved

$$C(rb \to rb) = \frac{1}{4}\sum_{a=1}^{8} \lambda_{11}^{a}\lambda_{33}^{a} = \frac{1}{4}(\lambda_{11}^{8}\lambda_{33}^{8})$$

$$= \frac{1}{4}\left(\frac{1}{\sqrt{3}}.\frac{-2}{\sqrt{3}}\right) = -\frac{1}{6}$$

Similarly $C(rb \to rb) = C(rg \to rg) = C(gr \to gr) = C(gb \to gb) = C(br \to br) = C(bg \to bg) = -\frac{1}{6}$

❸ Configurations where quarks swap colours e.g. $rg \to gr$



- Only matrices with non-zero entries in **12** and **21** position are involved

Gluons $r\bar{g}, g\bar{r}$

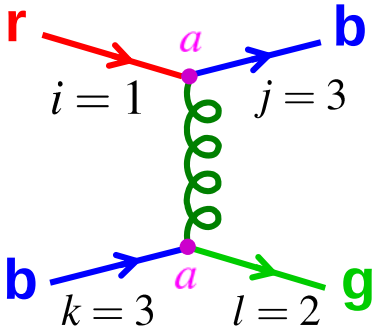
$$C(rg \to gr) = \frac{1}{4}\sum_{a=1}^{8} \lambda_{21}^{a}\lambda_{12}^{a} = \frac{1}{4}(\lambda_{21}^{1}\lambda_{12}^{1} + \lambda_{21}^{2}\lambda_{12}^{2})$$

$$= \frac{1}{4}(i(-i)+1) = \frac{1}{2}$$

$\hat{T}_{+}^{(ij)}\hat{T}_{-}^{(kl)}$

$$C(rb \to br) = C(rg \to gr) = C(gr \to rg) = C(gb \to bg) = C(br \to rb) = C(bg \to gb) = \frac{1}{2}$$

❹ Configurations involving 3 colours e.g. $rb \to bg$



- Only matrices with non-zero entries in the **13** and **32** position
- But none of the λ matrices have non-zero entries in the **13** and **32** positions. Hence the colour factor is zero

★ colour is conserved

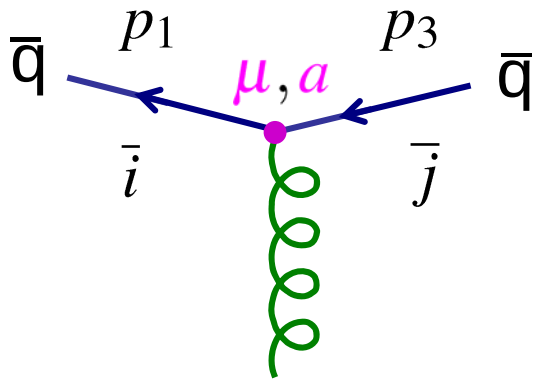
Colour Factors : Quarks vs Anti-Quarks

- **Recall the colour part of wave-function:**

$$r = c_1 = \begin{pmatrix}1\\0\\0\end{pmatrix} \quad g = c_2 = \begin{pmatrix}0\\1\\0\end{pmatrix} \quad b = c_3 = \begin{pmatrix}0\\0\\1\end{pmatrix}$$

- **The QCD qqg vertex was written:**

$$\bar{u}(p_3)c_j^{\dagger}\{-\tfrac{1}{2}ig_s\lambda^a\gamma^\mu\}c_i u(p_1)$$



★**Now consider the anti-quark vertex**

- **The QCD $\bar{q}\bar{q}g$ vertex is:**

$$\bar{v}(p_1)c_i^{\dagger}\{-\tfrac{1}{2}ig_s\lambda^a\gamma^\mu\}c_j v(p_3)$$

Note that the incoming anti-particle now enters on the LHS of the expression

- **For which the colour part is**

$$c_i^{\dagger}\lambda^a c_j = c_i^{\dagger}\begin{pmatrix}\lambda^a_{1j}\\ \lambda^a_{2j}\\ \lambda^a_{3j}\end{pmatrix} = \lambda^a_{ij}$$

i.e indices *ij* are swapped with respect to the quark case

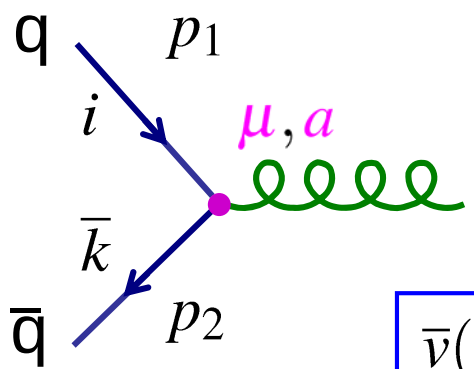
- **Hence**

$$\bar{v}(p_1)c_i^{\dagger}\{-\tfrac{1}{2}ig_s\lambda^a\gamma^\mu\}c_j v(p_3) \equiv \bar{v}(p_1)\{-\tfrac{1}{2}ig_s\lambda^a_{ij}\gamma^\mu\}v(p_3)$$

- **c.f. the quark - gluon QCD interaction**

$$\bar{u}(p_3)c_j^{\dagger}\{-\tfrac{1}{2}ig_s\lambda^a\gamma^\mu\}c_i u(p_1) \equiv \bar{u}(p_3)\{-\tfrac{1}{2}ig_s\lambda^a_{ji}\gamma^\mu\}u(p_1)$$

★**Finally we can consider the quark – anti-quark annihilation**

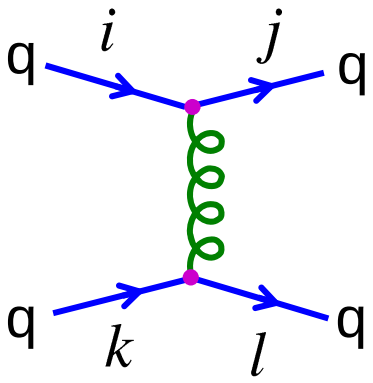


QCD vertex: $\bar{v}(p_2)c_k^\dagger\{-\frac{1}{2}ig_s\lambda^a\gamma^\mu\}c_i u(p_1)$

with $c_k^\dagger\lambda^a c_i = \lambda^a_{ki}$

$$\bar{v}(p_2)c_k^\dagger\{-\tfrac{1}{2}ig_s\lambda^a\gamma^\mu\}c_i u(p_1) \equiv \bar{v}(p_2)\{-\tfrac{1}{2}ig_s\lambda^a_{ki}\gamma^\mu\}u(p_1)$$

- **Consequently the colour factors for the different diagrams are:**



$$C(ik \to jl) \equiv \frac{1}{4}\sum_{a=1}^{8} \lambda^a_{ji}\lambda^a_{lk}$$

e.g.

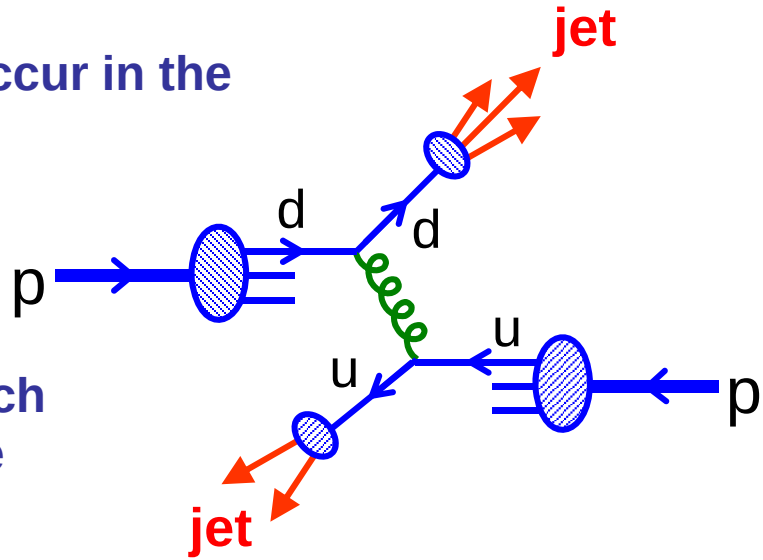
$$C(rr \to rr) = \tfrac{1}{3}$$
$$C(rg \to rg) = -\tfrac{1}{6}$$
$$C(rg \to gr) = \tfrac{1}{2}$$

Exercise:

Sum over all cc→cc the possibilities that fit the three forms above, and weight each by the number of those cases to get the total averaged/summed color factor.

Quark-Quark Scattering

- Consider the process $u + d \to u + d$ which can occur in the high energy proton-proton scattering
- There are nine possible colour configurations of the colliding quarks which are all equally likely.
- Need to determine the average matrix element which is the sum over all possible colours divided by the number of possible initial colour states



$$\langle |M_{fi}|^2 \rangle = \frac{1}{3} \cdot \frac{1}{3} \sum_{i,j,k,l=1}^{3} |M_{fi}(ij \to kl)|^2$$

- The colour average matrix element contains the average colour factor

$$\langle |C|^2 \rangle = \frac{1}{9} \sum_{i,j,k,l=1}^{3} |C(ij \to kl)|^2$$

- For $qq \to qq$

rr→rr,.. **rb→rb,..** **rb→br,..**

$$\langle |C|^2 \rangle = \frac{1}{9}\left[3 \times \left(\frac{1}{3}\right)^2 + 6 \times \left(-\frac{1}{6}\right)^2 + 6 \times \left(\frac{1}{2}\right)^2\right] = \frac{2}{9}$$

- **Previously derived the Lorentz Invariant cross section for $e^-\mu^- \to e^-\mu^-$ elastic scattering in the ultra-relativistic limit (handout 6).**

QED

$$\frac{d\sigma}{dq^2} = \frac{2\pi\alpha^2}{q^4}\left[1+\left(1+\frac{q^2}{s}\right)^2\right]$$

- **For ud $\to$ ud in QCD replace $\alpha \to \alpha_s$ and multiply by $\langle|C|^2\rangle$**

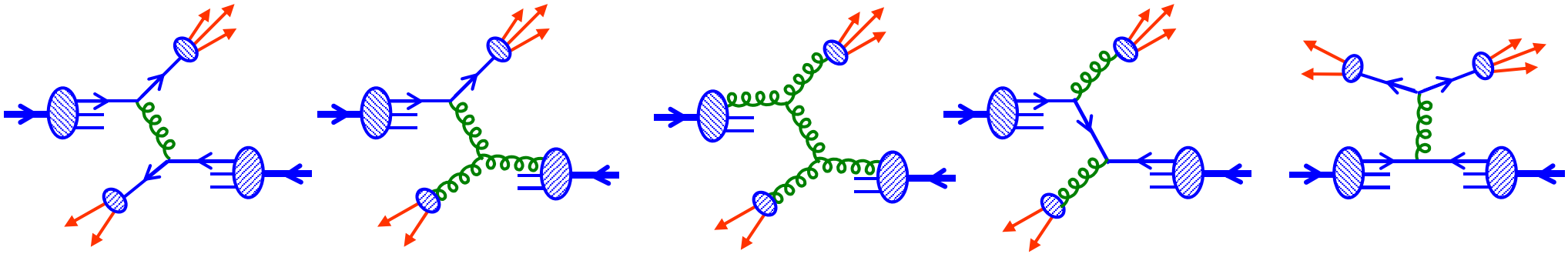
QCD

$$\frac{d\sigma}{dq^2} = \frac{2}{9}\frac{2\pi\alpha_S^2}{q^4}\left[1+\left(1+\frac{q^2}{\hat{s}}\right)^2\right]$$

Never see colour, but enters through colour factors. Can tell QCD is SU(3)

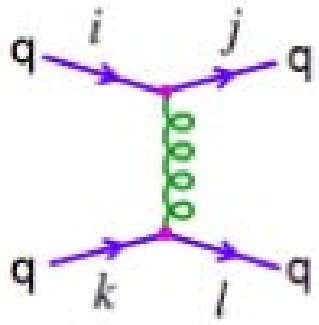
- **Here $\hat{s}$ is the centre-of-mass energy of the quark-quark collision**
- **The calculation of hadron-hadron scattering is very involved, need to include parton structure functions and include all possible interactions**

e.g. two jet production in proton-antiproton collisions



Exercise

- Consequently the colour factors for the different diagrams are:



$$C(ik \to jl) \equiv \frac{1}{4}\sum_{a=1}^{8} \lambda_{ji}^{a}\lambda_{lk}^{a}$$

e.g.

$$C(rr \to rr) = \tfrac{1}{3}$$
$$C(rg \to rg) = -\tfrac{1}{6}$$
$$C(rg \to gr) = \tfrac{1}{2}$$

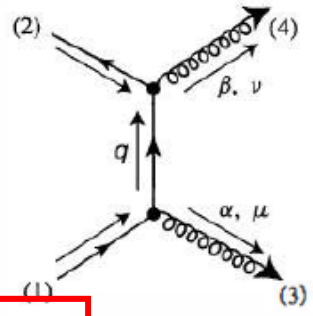
Exercise:

Sum over all cc→cc the possibilities that fit the three forms above, and weight each by the number of those cases to get the total averaged/summed color factor.

Gluon Color factor cases

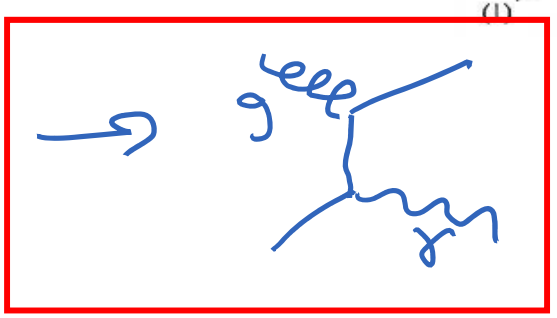
- qqbar

$$\mathcal{M}_1 = \frac{-g_s^2}{8}\frac{1}{p_1 \cdot p_3}\left\{\bar{v}(2)[\not\epsilon_4(\not p_1 - \not p_3 + mc)\not\epsilon_3]u(1)\right\} a_3^\alpha a_4^\beta (c_2^\dagger \lambda^\beta \lambda^\alpha c_1)$$



- Gluon Compton
  - Start small





10.4 The Gluon Emission Cross Section

We are now ready to include the gluon emission diagrams of Fig. 10.2. To calculate $\gamma^* q \to qg$, we take over our results for Compton scattering. In Section 6.14, we evaluated the closely related QED process $\gamma^* e \to \gamma e$ and found

$$\overline{|\mathcal{M}|^2} = 32\pi^2\alpha^2\left(-\frac{u}{s} - \frac{s}{u} + \frac{2tQ^2}{su}\right), \qquad (10.16)$$

Particles and Nuclei 2

- 4/7/2026
- Announcements:

- HW 5 finalized due date one week from today

- Reading: see email from Sat 4/4

$$\overline{|\mathcal{M}|^2} = 32\pi^2\left(e_i^2\alpha\alpha_s\right)\frac{4}{3}\left(-\frac{\hat{t}}{\hat{s}} - \frac{\hat{s}}{\hat{t}} + \frac{2\hat{u}Q^2}{\hat{s}\hat{t}}\right). \qquad (10.17)$$

The invariant variables are denoted $\hat{s}$, $\hat{t}$, $\hat{u}$ to indicate that we are considering a parton subprocess.

The factor $\frac{4}{3}$ takes into account the summation/averaging over final/initial colors. It can readily be deduced from counting the color lines shown in Fig. 10.6. Each color line can take one of three possible colors, so there are $(3 \times 3 - 1)$ different configurations, if we subtract the color singlet configuration as explained in Section 2.15. We now average over the three initial quark colors and obtain $\frac{8}{3}$. However, this result has to be divided by a factor 2 because of the unfortunate historical definition of α_s (see Section 2.15).

Most simple QCD diagrams are exactly analogous to QED diagrams, and the QCD cross section is obtained by the replacement

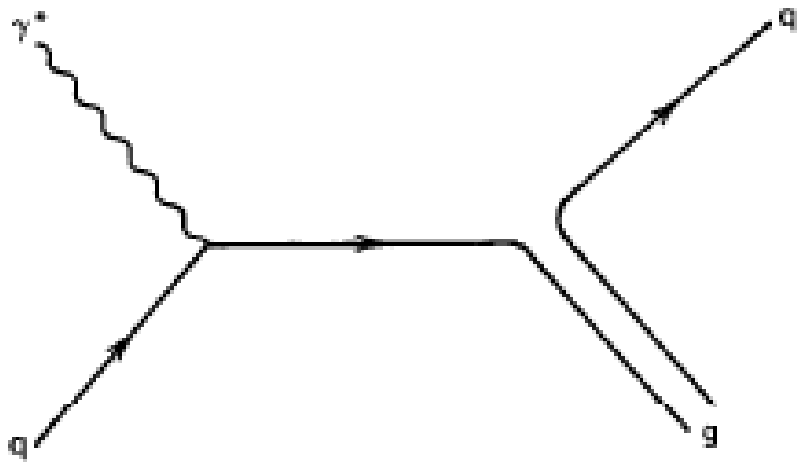


Fig. 10.6 Color lines for $\gamma^* q \to qg$.

your fingers" and divide the result by 2^n. This is why we divided the color factor $\frac{8}{3}$ of Fig. 10.6 by 2^1.

H&M Various Compton Processes

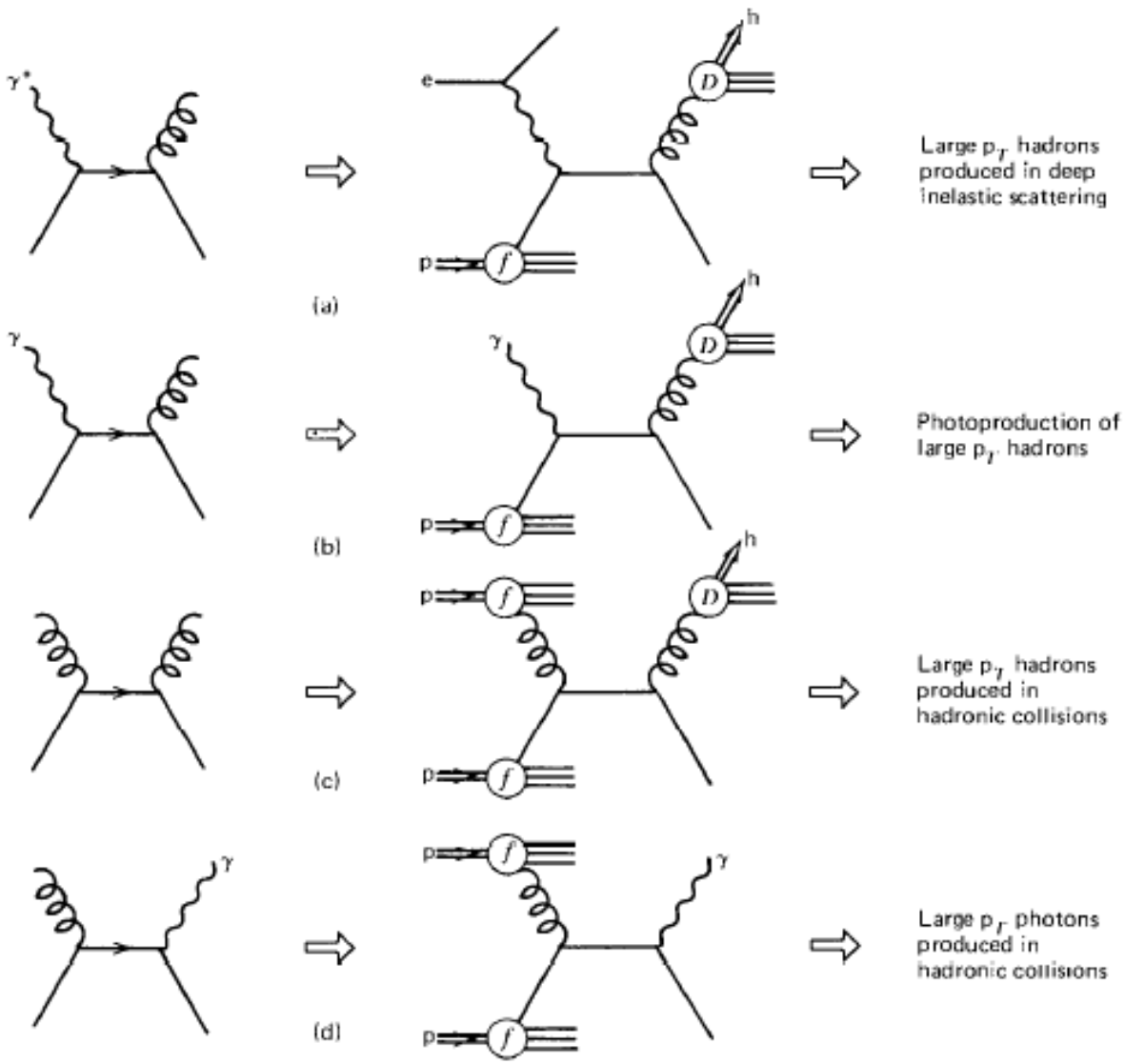


Fig. 11.13 Various observable processes which contain γ (or gluon)–quark Compton scattering, or the crossed reaction, as a parton subprocess; *f* and *D* denote structure and

- help for color factors for gluon here to be discussed later

$$\overline{|\mathcal{M}|^2} = 32\pi^2\left(e_i^2\alpha\alpha_s\right)\frac{4}{3}\left(-\frac{\hat{t}}{\hat{s}} - \frac{\hat{s}}{\hat{t}} + \frac{2\hat{u}Q^2}{\hat{s}\hat{t}}\right). \tag{10.17}$$

The invariant variables are denoted $\hat{s}, \hat{t}, \hat{u}$ to indicate that we are considering a parton subprocess.

The factor $\frac{4}{3}$ takes into account the summation/averaging over final/initial colors. It can readily be deduced from counting the color lines shown in Fig. 10.6. Each color line can take one of three possible colors, so there are $(3 \times 3 - 1)$ different configurations, if we subtract the color singlet configuration as explained in Section 2.15. We now average over the three initial quark colors and obtain $\frac{8}{3}$. However, this result has to be divided by a factor 2 because of the unfortunate historical definition of α_s (see Section 2.15).

Most simple QCD diagrams are exactly analogous to QED diagrams, and the QCD cross section is obtained by the replacement

$$\alpha^n \to C_F\alpha_s^n, \tag{10.18}$$

where n is the number of quark–gluon (or gluon–gluon) vertices in the diagram. The color factor C_F is obtained by simply summing and averaging in much the same way as we do for spin. However, without changing the cross section, the color factor can be altered by a redefinition of α_s. For example,

$$\alpha_s^n C_F \to (2\alpha_s)^n\left(C_F/2^n\right) = \left(\alpha_s'\right)^n C_F'. \tag{10.19}$$

The conventional definition of α_s corresponds to the second convention in (10.19). The final rule to obtain the conventional color factor is simple: "count color on your fingers" and divide the result by 2^n. This is why we divided the color factor $\frac{8}{3}$ of Fig. 10.6 by 2^1.

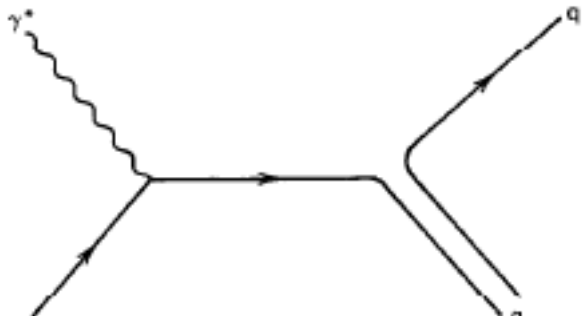


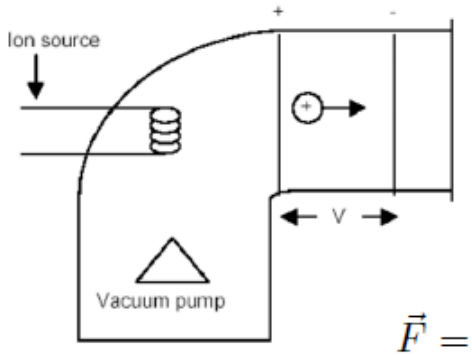
Fig. 10.6. Color lines for $\gamma^* q \to qg$

Detectors

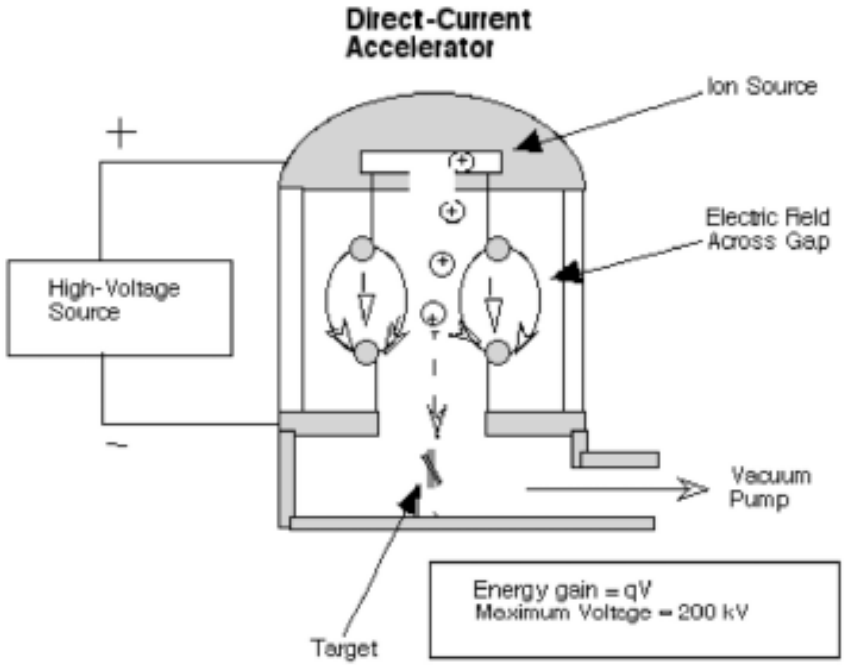
- Two important kinds tracking and at high energy calorimeters
  - EM cals most common.

Accelerators

- Voltage:
- Initial Stage from Rest



$$\vec{F} = q\vec{E} \Rightarrow \mathcal{E} = \left|\vec{F}\right| d = q \left|\vec{E}\right| d = qV$$



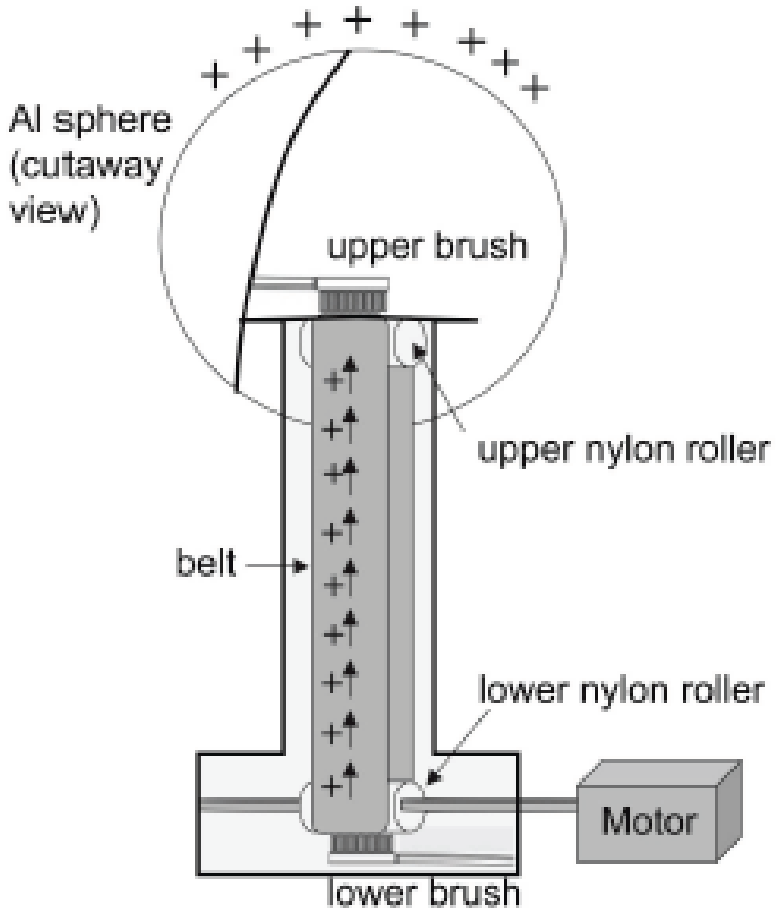
- Cockcroft Walton
- Max 200 kV
  - 200 keV

FIGURE 7.2
A Cockcroft-Walton Accelerator. Diagram courtesy of the Contemporary Physics Education Project (used with permission).

Van De Graf

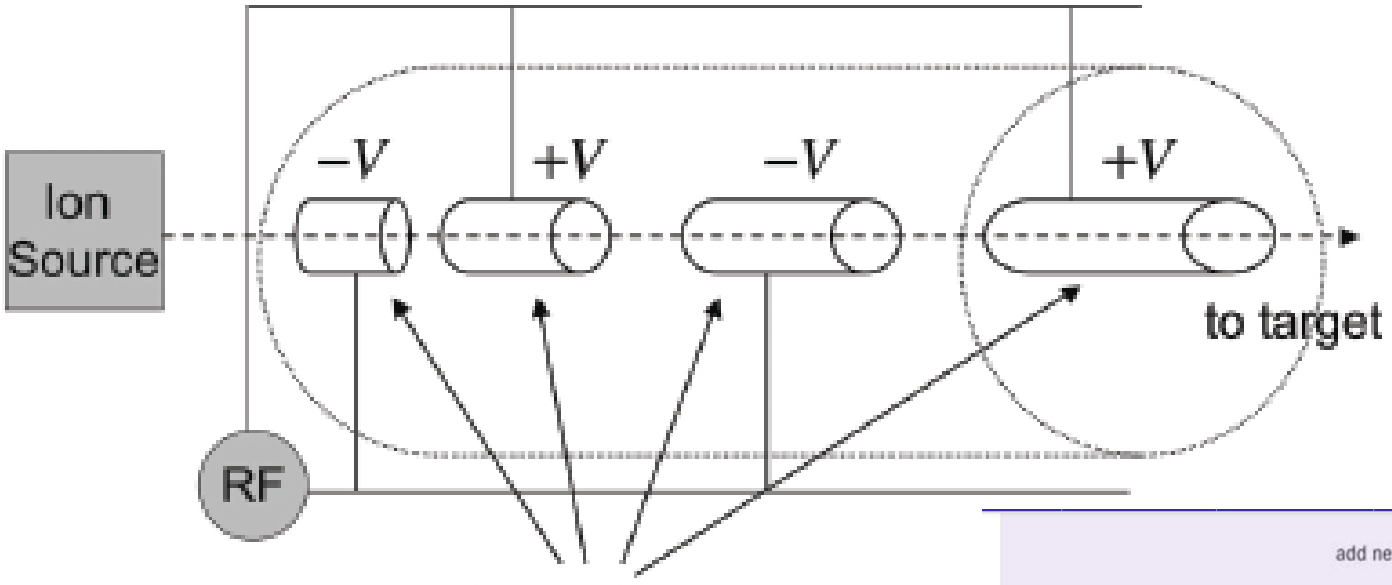
- Edwards !
- Build up charge/V
- Max 5-10 MV → MeV

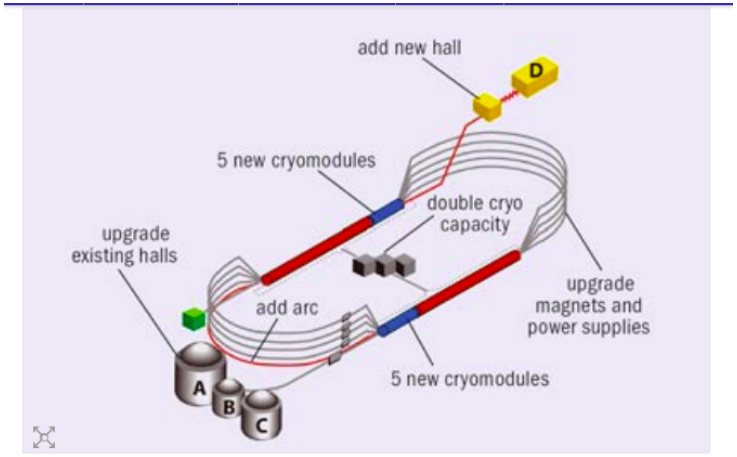
- Trick to reaccel
- 20-30 MeV



LINAC

- Linear Accelerator
- RF Frequency → E field

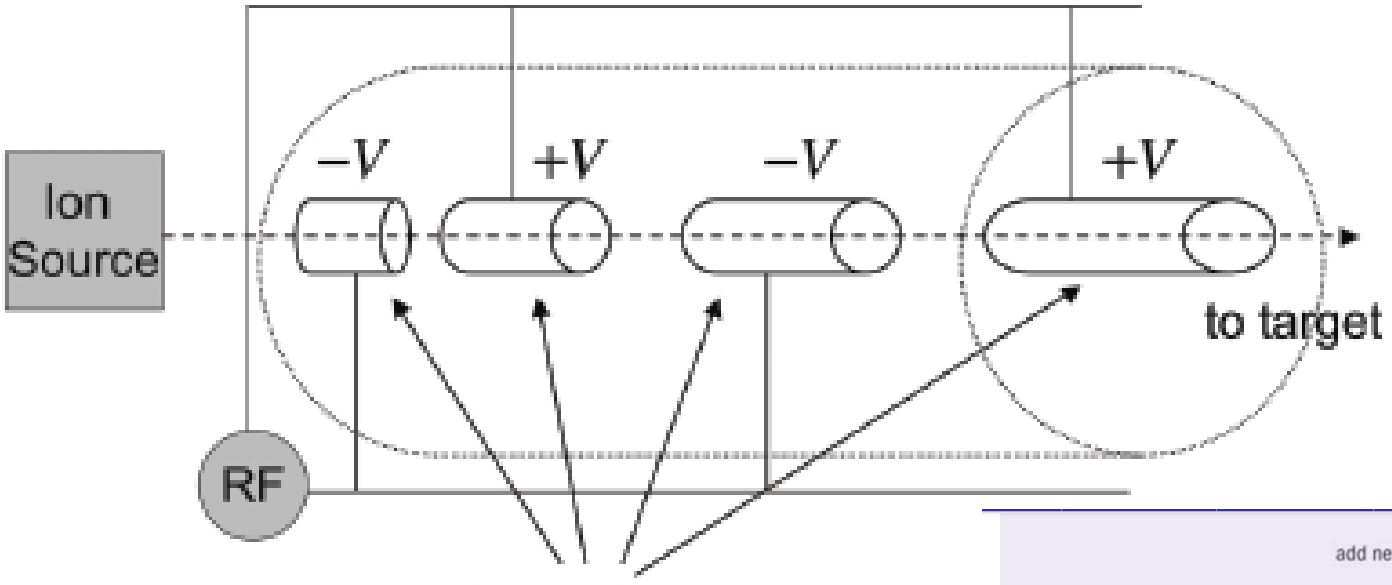


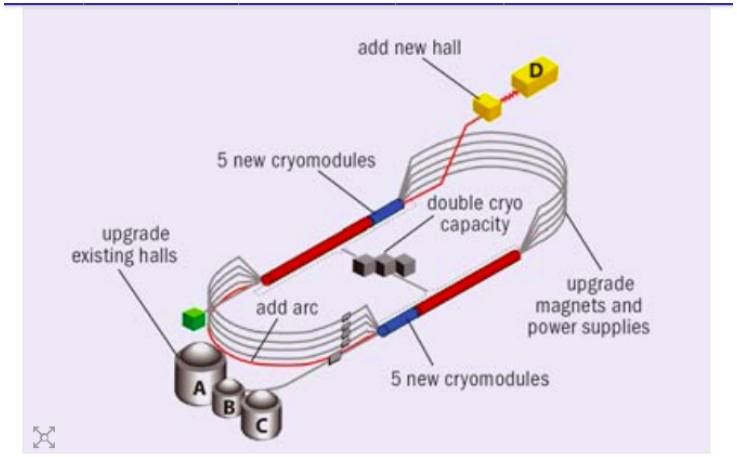


Schematic of the CEBAF configuration at Jefferson Lab, illustrating components of the 12 GeV

LINAC

- Linear Accelerator
- RF Frequency → E field



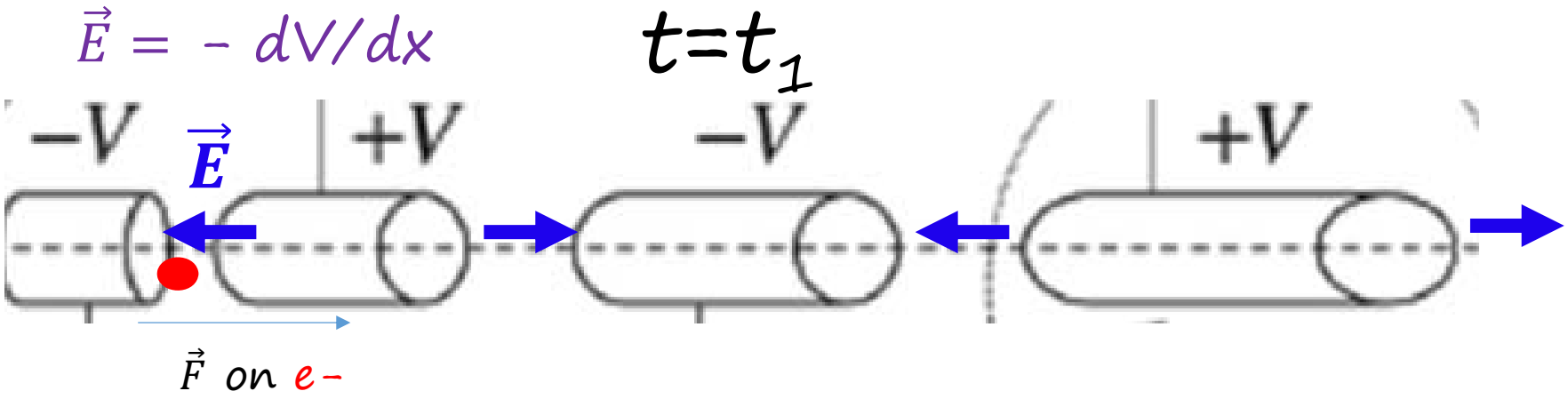


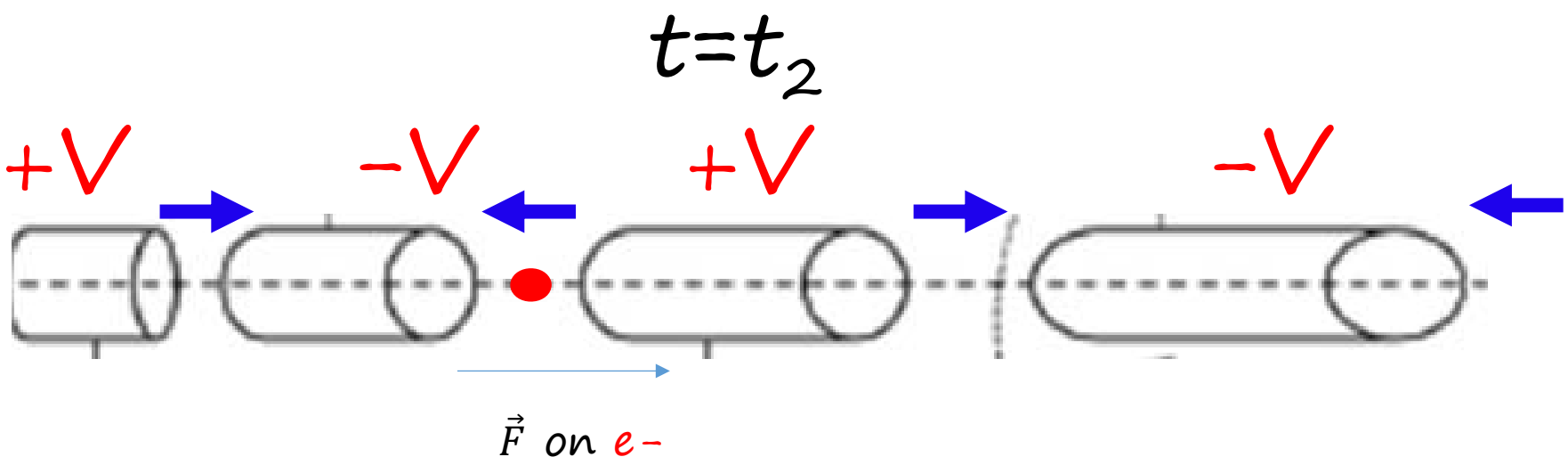
Schematic of the CEBAF configuration at Jefferson Lab, illustrating components of the 12 GeV

Q's about LINAC

- As described by Mann:

$\vec{E} = -dV/dx$



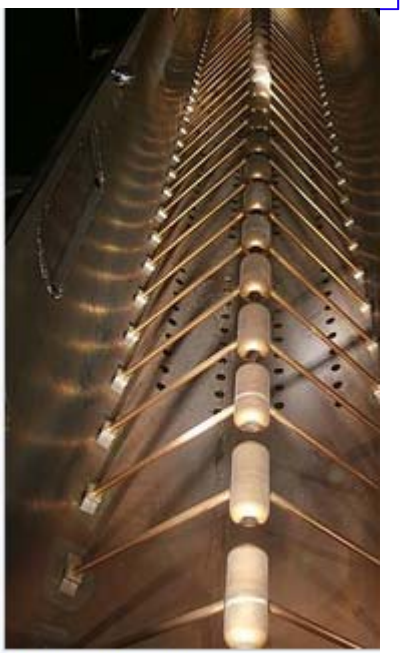


More about LINACS

From Wikipedia: LINAC

This type of linac was limited by the voltage sources that were available at the time, and it was not until after World War II that Luis Alvarez was able to use newly developed high frequency oscillators to design the first resonant cavity drift tube linac. An Alvarez linac differs from the Wideroe type in that the RF power is applied to the entire resonant chamber through which the particle travels, and the central tubes are only used to shield the particles during the decelerating portion of the oscillator's phase. Using this approach to acceleration meant that Alvarez's first linac was able to achieve proton energies of 31.5 MeV in 1947, the highest that had ever been reached at the time.[7]

- We can imagine the design of the "cavities" can greatly affect the performance
  - local non uniformities of the field deflecting/"defocusing" beam
- Alvarez Design: allow some fields in "resonant" cavities → better acceleration-- still exclude field though when not optimal
- There are probably an ~endless chain of improvements than can be had by more and more clever cavity designs
- This is generally true for nearly all components of accelerators – **"Accelerator Physics"** is still a thing ie a subfield of physics – PhD's from select group of universities with partnerships with BNL, CERN, etc.



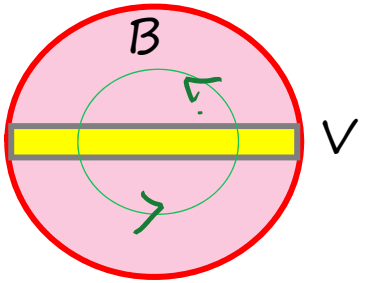
Alvarez type linac

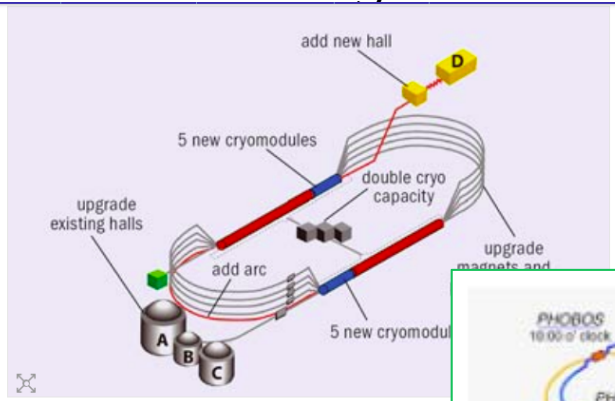
Alvarez: and other Names

- Luis Alvarez was a very famous experimentalist that touched many fields besides just physics
  - (Nobel Prize-bubble chamber, Manhattan project, etc.)
  - Him and his son an geologist named Walter A. discovered and deduced the KT boundary – geological layers where the meteorite that caused the Dinsosaur extinction – nuclear physics of the elemental analysis
  - He famously built cosmic ray based detectors to look inside large objects like the Egyptian pyramids
- **Names:** there is a history of recalling some of these big names
- I've met several older colleagues who are "name droppers", they met or worked with people like Alvarez.
- Lederman/Schwartz $\nu\mu$ – I met both of these guys –they were great
- **Often their histories are a bit problematic and it's good to acknowledge this**
  - I've heard Alvarez was especially arrogant, a common problem
- A couple points on that:
- The "Pantheon" reflects the pervasive sexism and implicit racisms of their times (though for latter pretty diverse for the times, Jewish/Asian/Alvarez) –
  - Photo libratry (~all male) pictures at BNL taken down
- **Einstein** : mostly seems nice – but bad model of celebrating the LONE MAD Scientist – modern work always involves lots of collaboration - social skills/respectful workplace - Mentionned Sam Ting
  - a professor I knew liked to tell the story of driving Einsein to a wedding
- Large number of early heros came from New York City – poor/rough neighborhoods especially the Bronx. Their "you-messin with me?!?!" argumentative styles, is a legacy can still be found today in some places in particle physics – lots of yelling, etc.
  - A thick skin still is needed, but we are working to make such interactions minimized or dissappear! and they are as we slowly get more diverse. **APS-IDEA!!!**

Synchotrons

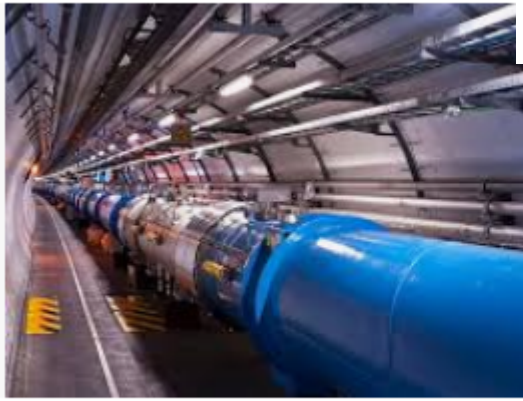
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  - Combinations of "LINAC"s and Magnet Trains



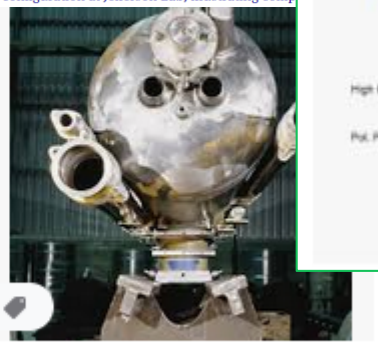


Schematic of the CEBAF configuration at Jefferson Lab, illustrating comp

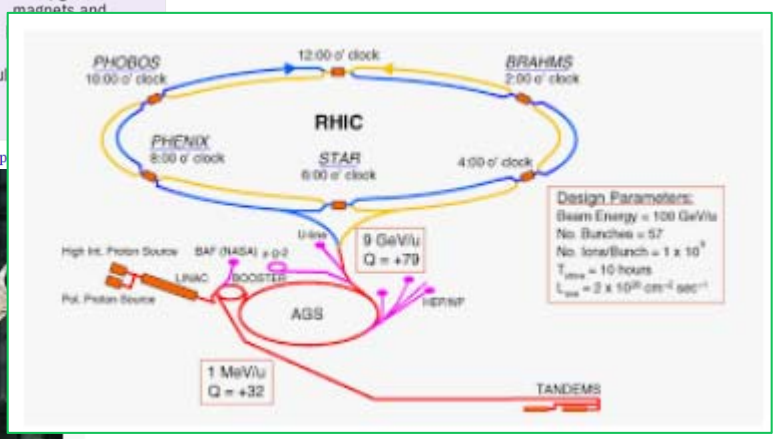
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Synchotron Considerations Resonances

- Magnetic Field -- better for bending / steering

$$\vec{F} = q\left(\vec{E} + \frac{\vec{v}}{c} \times \vec{B}\right) = \frac{q}{c}\vec{v} \times \vec{B}$$

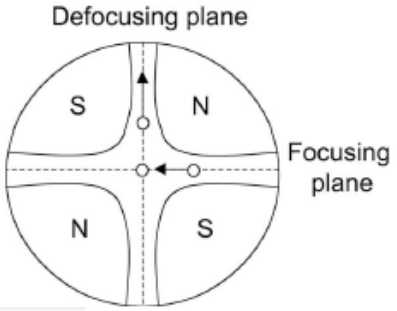
- Bending Magnets Dipole

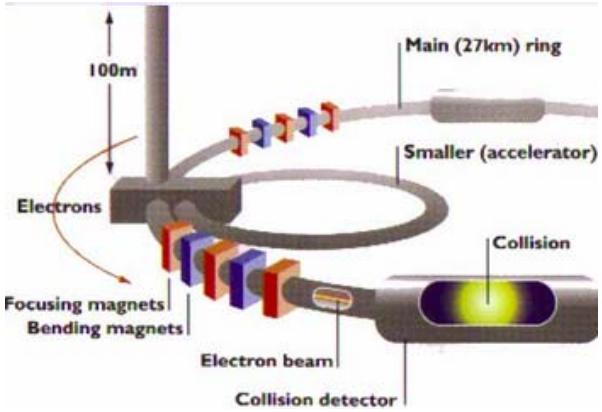
$$Br = \frac{pc}{q} :$$

- Focusing : Quadropole only one plane

- Trains alternating these "FODO

Quad





H&M Various Compton Processes

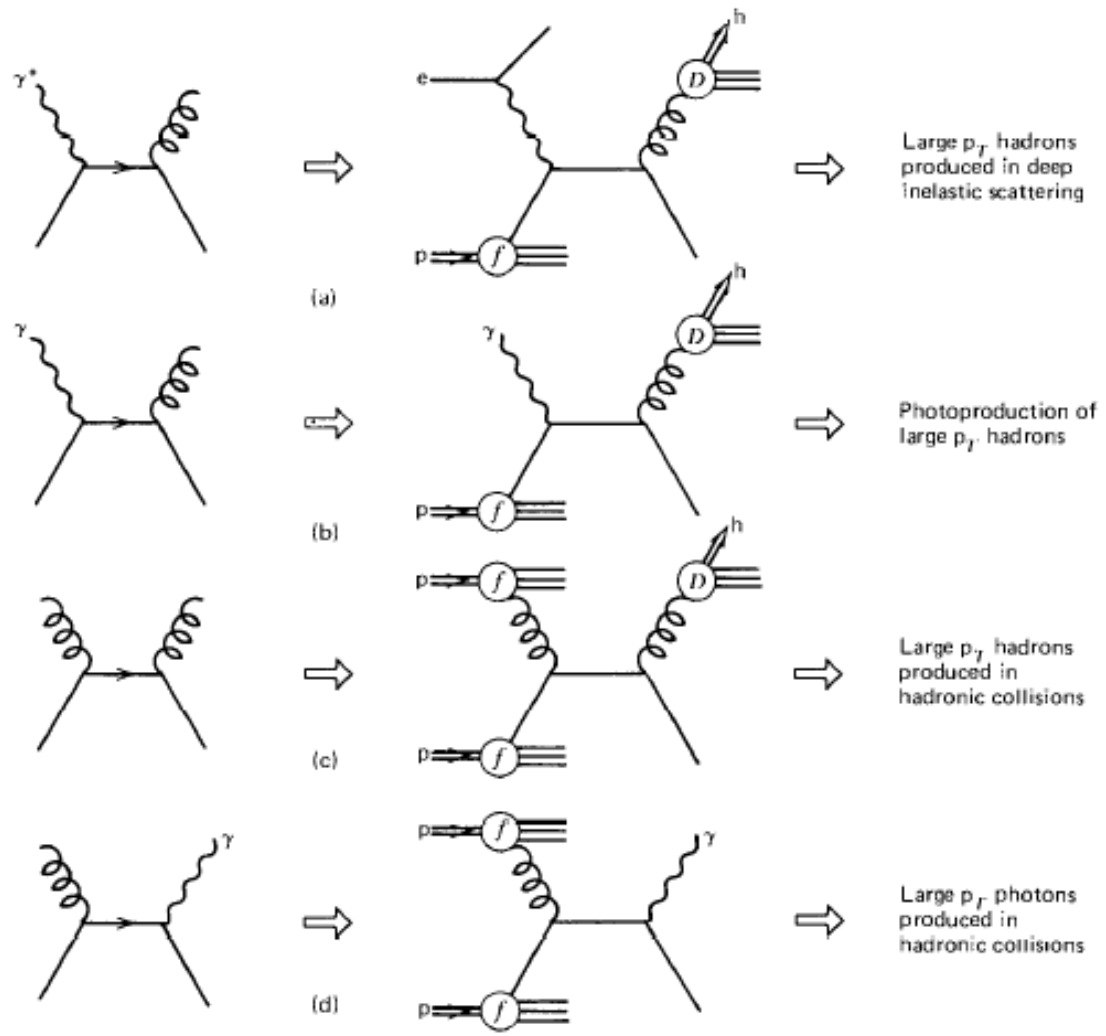
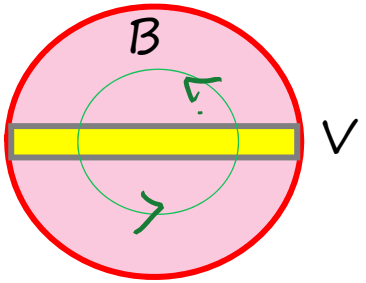
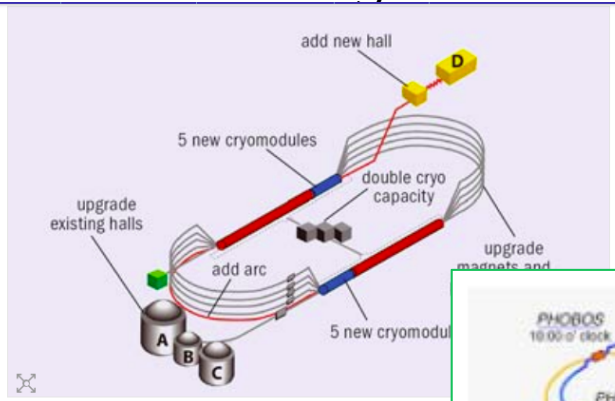


Fig. 11.13 Various observable processes which contain γ (or gluon)–quark Compton scattering, or the crossed reaction, as a parton subprocess; f and D denote structure and fragmentation functions, respectively.

Synchotrons

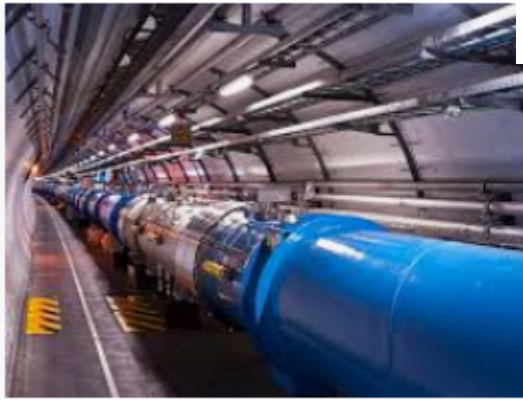
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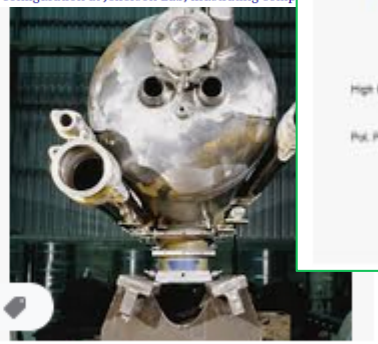


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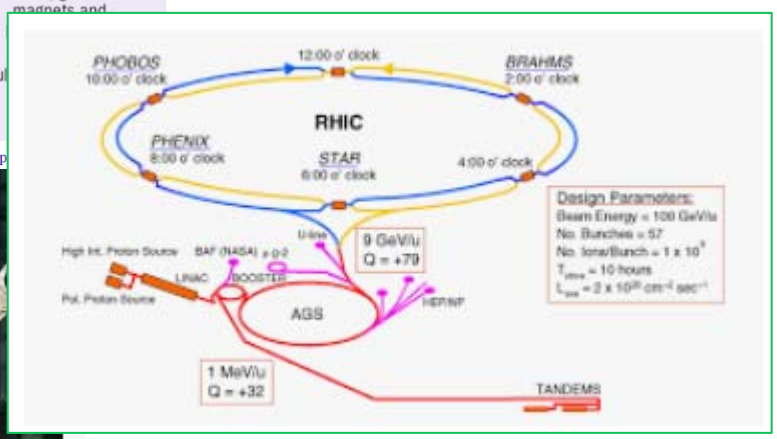
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SYNCHotron

- Obviously the rate at which linacs acc must match increase in Bending Bfields

V acc oscillations driven b

$$\Omega = \frac{2\pi}{T} = \frac{pc^2}{\mathcal{E}R}$$

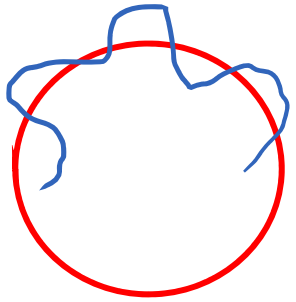
RHIC
T = 6 μs
fyi timing 10's of ns oscillations

Linac sections must have special ω → k is integer

$$\omega = k\Omega = k\frac{pc^2}{\mathcal{E}R} \rightarrow \frac{kc}{R}$$

- See transition energy next slide

- Focusing related to oscillations for slightly different orbits
- Betatron Oscillations

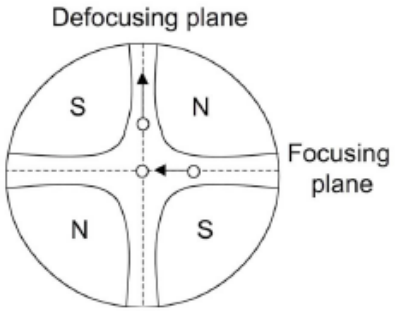


- Also oscillations in beam direction:
- **Synchotron Oscillations BUNCHING**

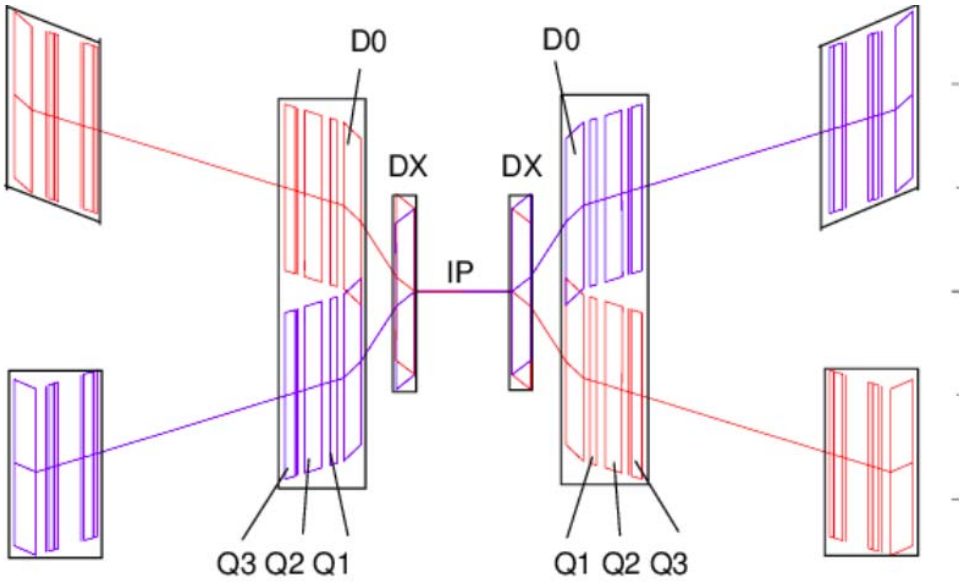
BEAM RESONANCES – IR's

- Betatron Osc. → 2 D + Sych Os 1D

- **RESONANCES !!! AVOID** -- TUNING PROCESS
  - Adjust magnet locations -- add
  - Later, adjust Vacc RF's , B



- This is also greatly affected by Interaction regions and "beam dumps"



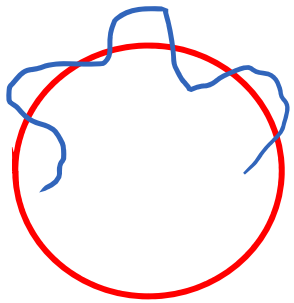
Exercise

- Show that ω can be written as an integer k times the revolution time.

$$\Omega = \frac{2\pi}{T} = \frac{pc^2}{\mathcal{E}R}$$

RHIC
T = 6 μs
fyi timing 10's of ns oscillations

$$\omega = k\Omega :$$



Particles and Nuclei 2

- 4/7/2026
- Announcements:

- HW 5 finalized due date one week from today

- Reading: see email from Sat 4/4

("Help") For gluon Compton color factor

- Part 1b) I will accept any answer **with ANY kind of physics-based logic/justification** for about 95% of the credit [for the color factor part/cross section part—all but ~0.5-1pt].
- For the remaining (~1 pt) credit, I will apply only a little subjectivity into the effort put into your justification (which doesn't have to be great or correct but should involve some reference to the general method of calculating CF's (#q/g, averaging etc.)
- I will discuss a few more details next time but for example
  - If you choose to do the easier color factor for q+photon→q+g or the reverse, make sure you choose a quark flavor and then also include the extra factor for the fractional charge of the quark (or use an average over all flavors—any averaging w/ reasonable justif is OK)
  - In that case I recommend for the color using the result in Halzen&Martin and explaining it based on the color counting method described there

  - For the gq→gq one option is to also use that color counting method [and don't worry if correct] whether this yields the correct answer or not
  - You can also just try to follow what Griffiths provides, I will not be very critical—this I will give a little more info next time

Read this!

- The method I'm referring to is described here:

(THIS IS IN HANDOUT p.40 of pdf, 211 of book)

$$\overline{|\mathfrak{M}|^2} = 32\pi^2\left(e_i^2\alpha\alpha_s\right)\frac{4}{3}\left(-\frac{\hat{t}}{\hat{s}} - \frac{\hat{s}}{\hat{t}} + \frac{2\hat{u}Q^2}{\hat{s}\hat{t}}\right). \tag{10.17}$$

The invariant variables are denoted $\hat{s}, \hat{t}, \hat{u}$ to indicate that we are considering a parton subprocess.

The factor $\frac{4}{3}$ takes into account the summation/averaging over final/initial colors. It can readily be deduced from counting the color lines shown in Fig. 10.6. Each color line can take one of three possible colors, so there are $(3 \times 3 - 1)$ different configurations, if we subtract the color singlet configuration as explained in Section 2.15. We now average over the three initial quark colors and obtain $\frac{8}{3}$. However, this result has to be divided by a factor 2 because of the unfortunate historical definition of α_s (see Section 2.15).

Most simple QCD diagrams are exactly analogous to QED diagrams, and the QCD cross section is obtained by the replacement

$$\alpha^n \to C_F\alpha_s^n, \tag{10.18}$$

where n is the number of quark–gluon (or gluon–gluon) vertices in the diagram. The color factor C_F is obtained by simply summing and averaging in much the same way as we do for spin. However, without changing the cross section, the color factor can be altered by a redefinition of α_s. For example,

$$\alpha_s^n C_F \to (2\alpha_s)^n (C_F/2^n) = (\alpha_s')^n C_F'. \tag{10.19}$$

The conventional definition of α_s corresponds to the second convention in (10.19). The final rule to obtain the conventional color factor is simple: "count color on your fingers" and divide the result by 2^n. This is why we divided the color factor $\frac{8}{3}$ of Fig. 10.6 by 2^1.

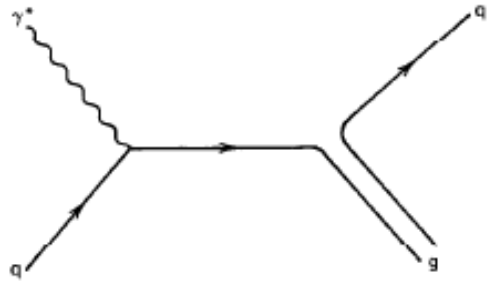


Fig. 10.6 Color lines for $\gamma^* q \to qg$.

$$\overline{|\mathcal{M}|^2} = 32\pi^2\left(e_i^2\alpha\alpha_s\right)\frac{4}{3}\left(-\frac{\hat{t}}{\hat{s}} - \frac{\hat{s}}{\hat{t}} + \frac{2\hat{u}Q^2}{\hat{s}\hat{t}}\right). \qquad (10.17)$$

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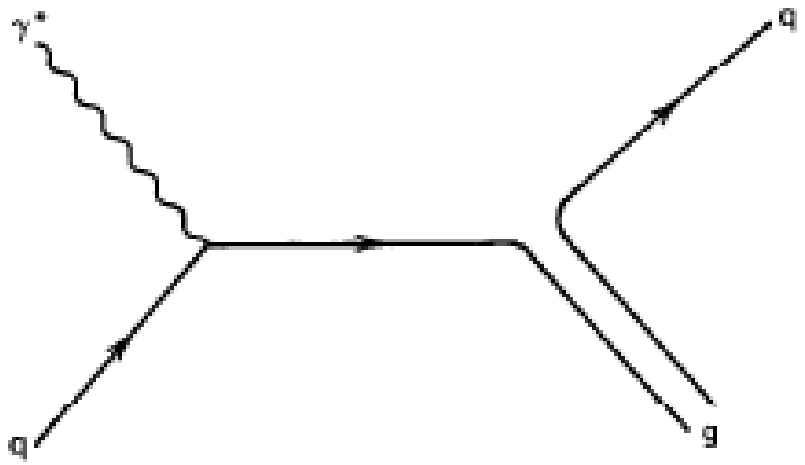
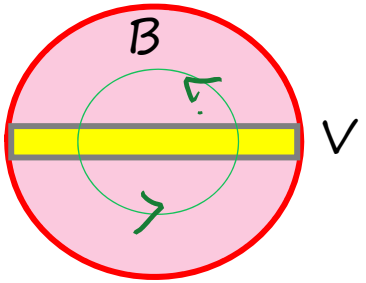


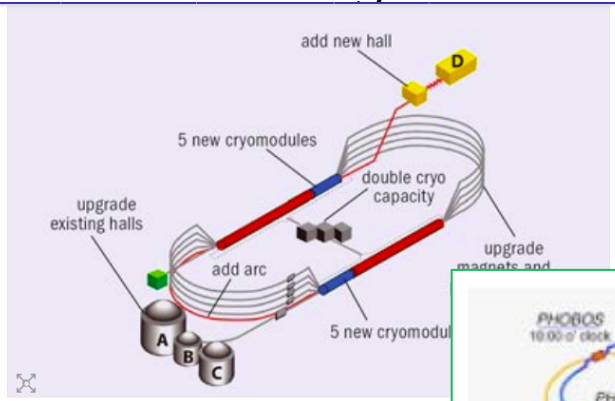
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Synchotrons

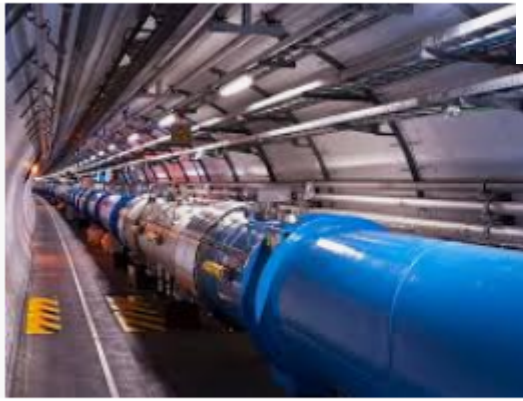
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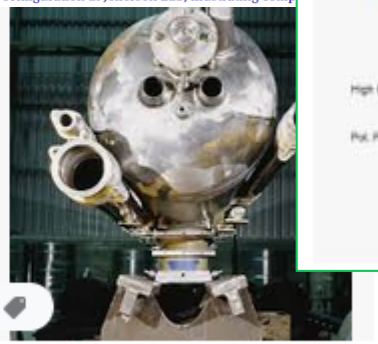


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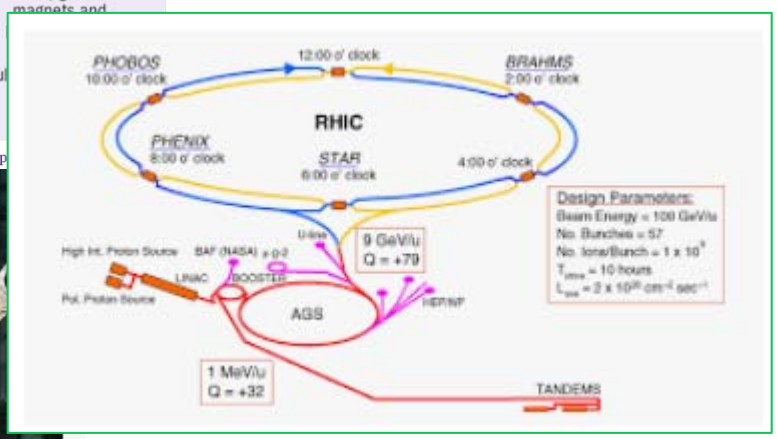
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SYNCHotron

- Obviously the rate at which linacs acc must match increase in Bending Bfields

V acc oscillations driven b

Linac sections must have special a

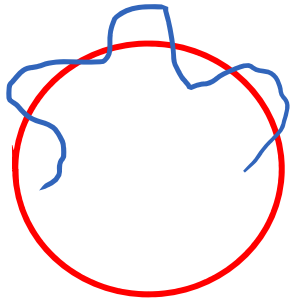
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RHIC
T = 6 μs
fyi timing 10's of ns oscillations

$$\omega = k\Omega = k\frac{pc^2}{\mathcal{E}R} \rightarrow \frac{kc}{R}$$

- See transition energy next slide

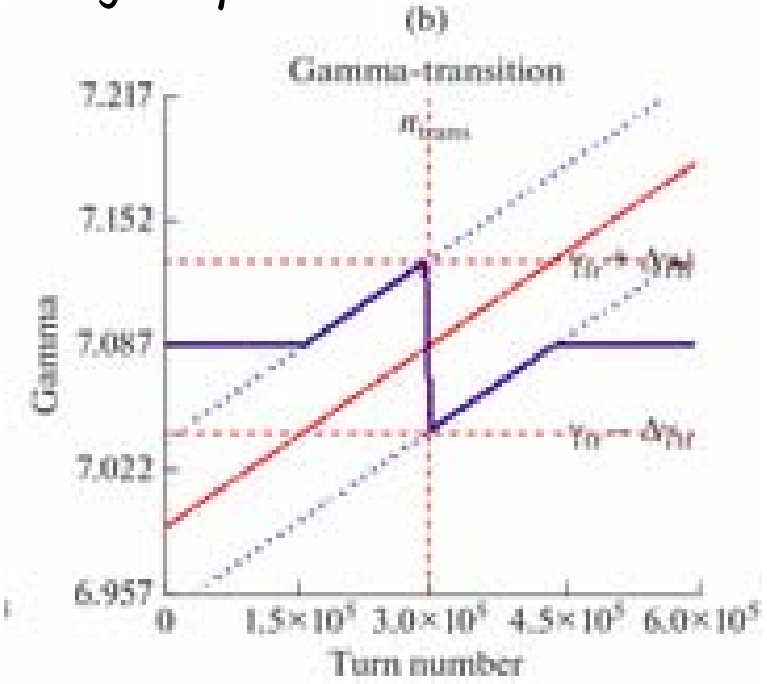
- Focusing related to oscillations for slightly different orbits
- Betatron Oscillations



- Also oscillations in beam direction:
- **Synchotron Oscillations BUNCHING**

Transition Energy !!!

- **Cool relativistic effect!** During acceleration in synch'trn:
- Transition is the critical energy where the synchrotron's revolution frequency of particles becomes *independent of momentum* Below this energy, faster particles have **higher frequency;** [cause they're increasing speed as normal]
- Above it, once c is reached , they have **lower frequency** due to longer path lengths
  - (speed same but > mom harder bend still)
- Requires a "phase jump" in the RF acceleration



Electron Vs Protons

- Synchotron Radiation – Bremstrahlung

Since electrons are lighter and so much easier to accelerate, why not build electron synchrotrons? This can be done, but there is a significant cost because all charged particles radiate energy as they accelerate. Circularly moving particles at higher and higher speeds accelerate more and more, losing energy ΔE with each revolution:

$$\Delta E = \frac{4\pi}{3}\frac{q^2\beta^3\gamma^4}{R} \longrightarrow \frac{4\pi}{3}\frac{q^2}{R}\left(\frac{E}{mc^2}\right)^4 \quad \text{as} \quad v \to c \tag{7.13}$$

where the formula for energy loss is given in Jackson (eq. (14.31)) [71]. Hence the energy loss varies inversely with the fourth power of the mass, and so

$$\frac{(\Delta E)_{\text{electron}}}{(\Delta E)_{\text{proton}}} = \left(\frac{m_{\text{proton}}}{m_{\text{electron}}}\right)^4 \simeq 10^{13} \tag{7.14}$$

which is a huge energy loss: at 20 GeV beam energy it is 16 MeV per turn.

- Advantages: Use electron rings as light sources NSLS (II) at BNL → SLAC for condensed matter



Storage Rings

- Sometimes in older language some Synchotrons can be called storage rings
- Nowadays usually refers to some unstable particle
- Muon Storage Ring g-2 experiment



Muon g-2 Storage Ring Move | Ten years ...
flickr.com



50-Foot-Wide Particle Storage Ring ...
bnl.gov

- Bigger ones have been proposed and probably even made
- FRIB to some extent

Flux, Luminosity Bunches

- The **focusing part (tranverse size)** determines the other most important part of an accelerator Flux/ LUMINOSITY
- **Smaller Area higher Flux/Lumi**

Flux: A beam of incoming pointlike particles has flux density j (#/m^2/sec) uniformly spread ove an area A_b [m^2] with total flux $J = jA$, [#/sec]. The flux density can be written $j = n_b v_b$, where

$$L = \frac{1}{\sigma}\frac{dN}{dt}.$$

- For colliders **Luminosity** ~same units but like effective flux taking into account both beam fluxs – x.s. in def unavoidable
- Bunching (part of focusing) big driver

$$\mathcal{R} = \sigma\mathcal{L} \quad \text{with } \mathcal{L} = \mathrm{f}n\frac{N_1 N_2}{A}$$

where the quantities associated with the luminosity $\mathcal{L}$ are

f = frequency of revolution of bunches
n = number of bunches per beam
N_1, N_2 = number of particles per bunch in beams 1,2
A = area of beam
σ = cross-section for interaction
(computed from underlying the

Usually given in "inverse" X barns

E.g. perhaps 1 fb^{-1} of data taken in one year

$\mathcal{L}_{1\,\mathrm{bcw}} \propto \frac{N}{A \cdot s}$

- Integrated Luminosity

$$L_{\mathrm{int}} = \int L\,dt.$$

Cutting Edge – Stochastic Cooling

- Put some electron beam near proton beam
- collisions with electrons on periphery of transverse beam "spot" can help focus and stabilize
- Somewhat like Laser cooling (lasers are probably used here)

- p+p cross section is 50 mb

- Inverse luminosity Example – 50 mb-1 of integrated luminosity of p+p events = 1 event

Calc num events from NL paper

- In example NL lunch paper: LHC data "data set corresponding to an inverse lumi of 138 fb^{-1} "

$138 \frac{1}{fb}$ $50\,mb =$

$\frac{1}{10^{15}}$

$\mathcal{L} \times \frac{d\sigma}{dp^3} = \frac{dN}{N_{ev}}$ $\quad \sigma_{tot}$

$\left(10^{15}(138)\frac{1}{b}\right) 50 \cdot 10^{-3}\, b$

$10^{15} \cdot 10^{2} \cdot 10^{1} \cdot 10^{-3}$

10^{15} events

spectra

Introduction: detector tasks

- Momentum measurement: by measuring the particle deflection in a magnetic field (eg: magnetic spectrometer)
  - Position measurement: localize hits of a charged particle (eg: wire chambers, segmented scintillators or calorimeters)
- Energy measurement: deposition of energy in a localized volume (eg: calorimeters)
- Particle identification: mass and charge of the particle (eg: Cerenkov detectors)
- Triggering: select events of interest
- Data acquisition system: readout of an event and storage after positive decision.

Introduction: detector performances

- Time
  - Response time: time which is required to produce a signal after the passage of a particle in the detector (from ~10 ns to ~100 μs).
  - Deadtime: time that must elapse following the passage of a particle before the detector is ready for the next particle.

- Efficiency

  That is (event registered)/(events emitted by source)

  The efficiency is the product of at least two quantities:
  - Intrisic efficiency: (event registered)/(events impinging on detector)
  - Geometric efficiency: (event impinging on detector)/(event emitted by source)

- Resolution/Accuracy

Scintillators

- MOST IMPORTANT DETECTOR TYPE IN Medium and High energy particle physics.

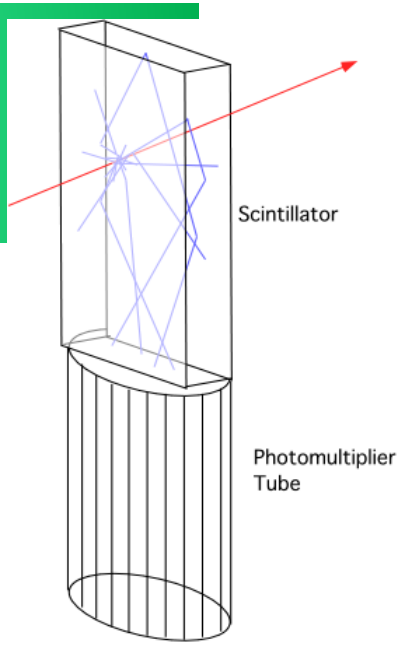
- **ALL PURPOSE TRIGGER!!!!**

- Translucent Material that emits visible or near visible light when charge particle passes through.
- Light is collected by light detection devices converted into voltage/current signal

- Mostly for triggering just yes/no.
  - (at low energy can give precise energy measurements)

- At high energy these can also provide the ~best timing resolution of all detectors, so also used for that (mainly for PID)
  - time → velocity → p → combined with E meas, PID.

Scintillators



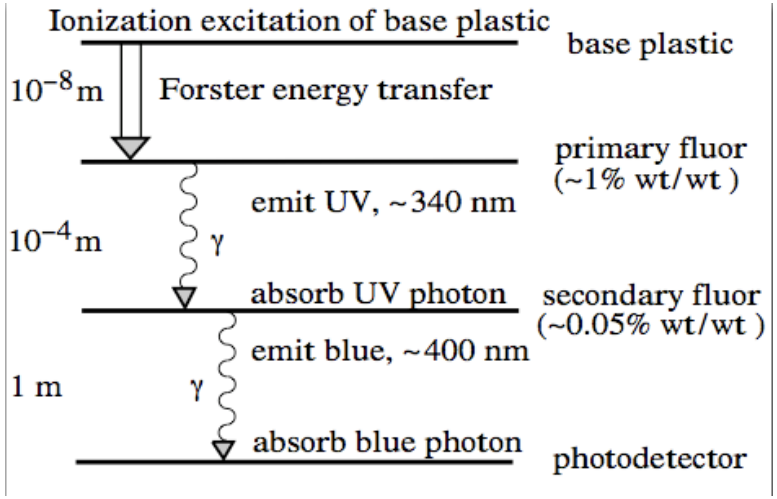
Scintillation Counters are probably the most widely used detectors in Nuclear and High Energy Physics.

Scintillator material are special material that

-emit light when traversed by energetic particles and

-can shift the wavelength of this light

to be harnessed by PMTs



They can be solid, liquid (even gas)

They can be molded in all kind of shapes

Example of position detectors performance

Table 28.1: Typical resolutions and deadtimes of common detectors. Revised September 2009.

Detector Type	Accuracy (rms)	Resolution Time	Dead Time
Bubble chamber	10–150 μm	1 ms	50 ms[a]
Streamer chamber	300 μm	2 μs	100 ms
Proportional chamber	50–100 μm[b,c,d]	2 ns	200 ns
Drift chamber	50–100 μm	2 ns[e]	100 ns
Scintillator	—	100 ps/n[f]	10 ns
Emulsion	1 μm	—	—
Liquid argon drift [7]	~175–450 μm	~ 200 ns	~ 2 μs
Micro-pattern gas detectors [8]	30–40 μm	< 10 ns	20 ns
Resistive plate chamber [9]	$\lesssim$10 μm	1–2 ns	—
Silicon strip	pitch/(3 to 7)[g]	*h*	*h*
Silicon pixel	2 μm[i]	*h*	*h*

Interaction of particles with matter

Charged particles:

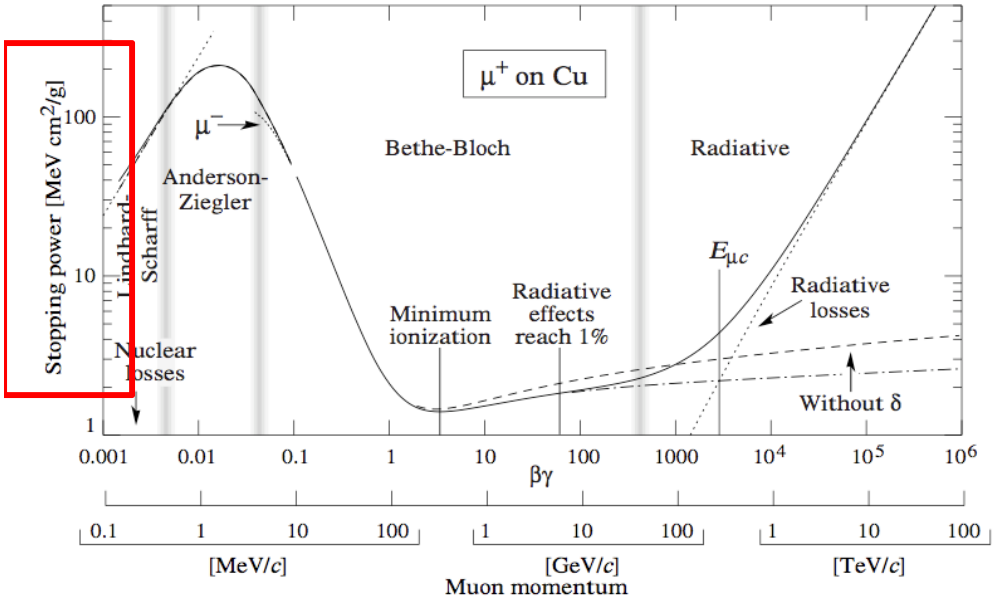
Energy and trajectories get degraded as they pass through matter:

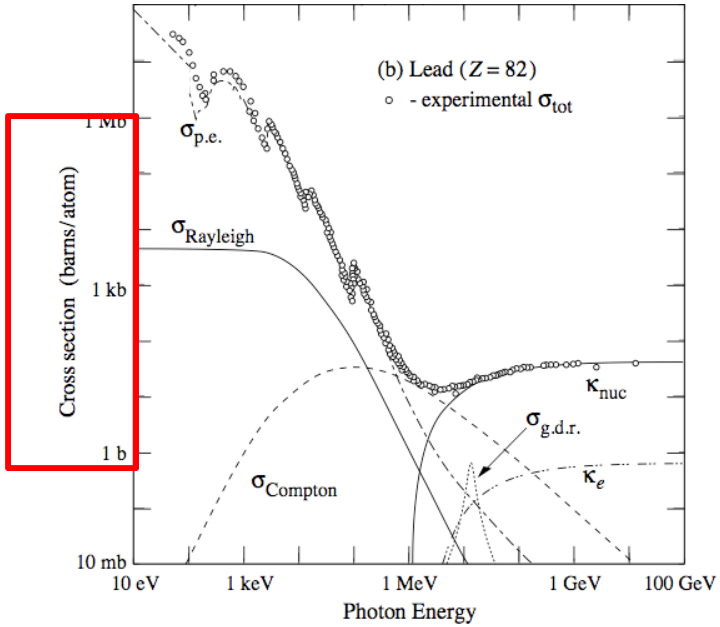
- Ionization
- Bremsstrahlung
- Cherenkov radiation

Photons:

Flux gets decreased as they go through matter. All or nothing interactions

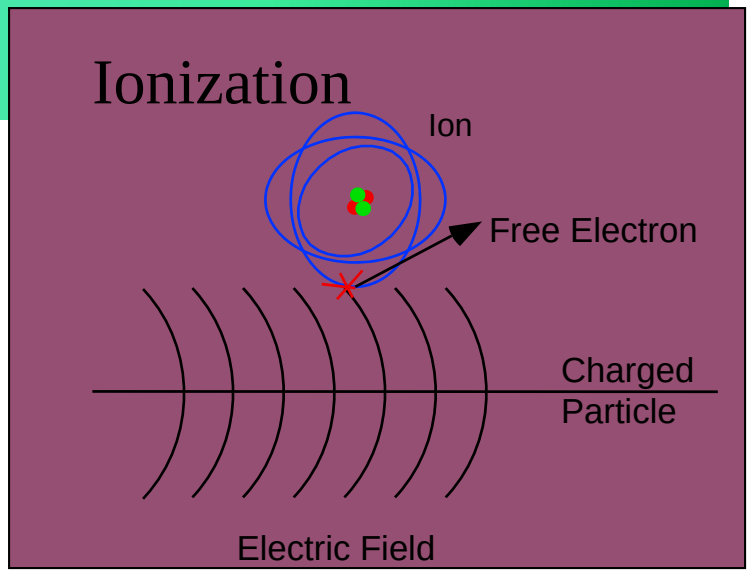
- Photo-electric effects
- Compton scattering
- Pair production

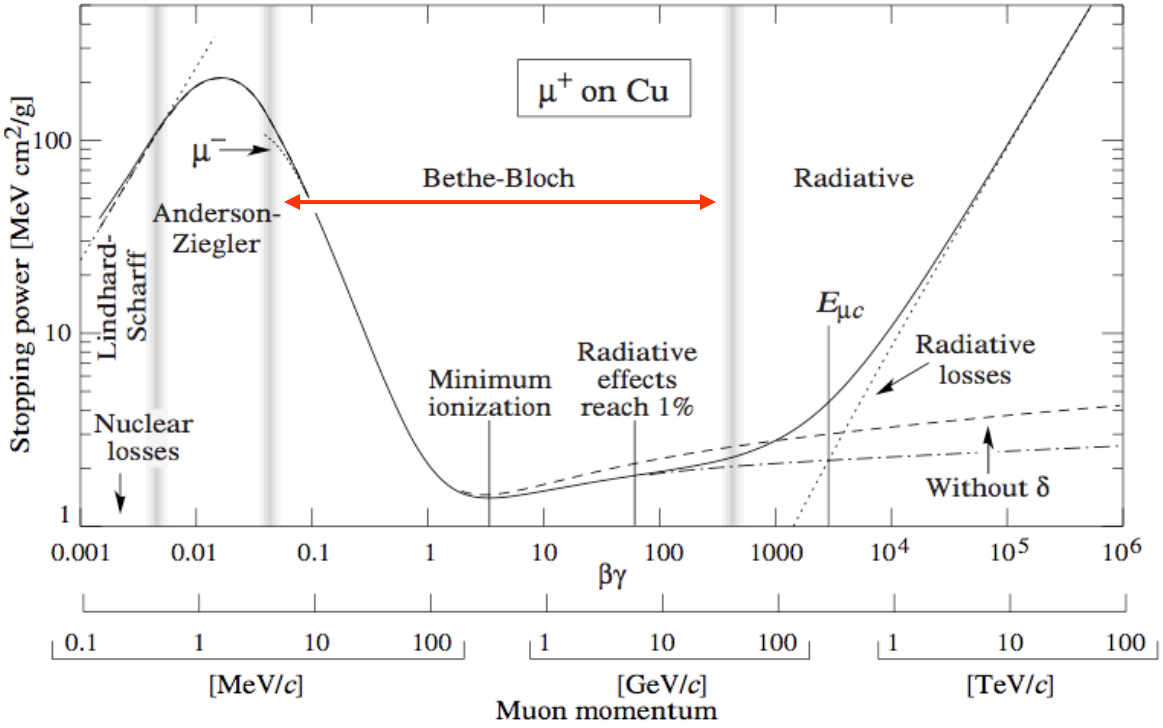




Charged particles- Ionization

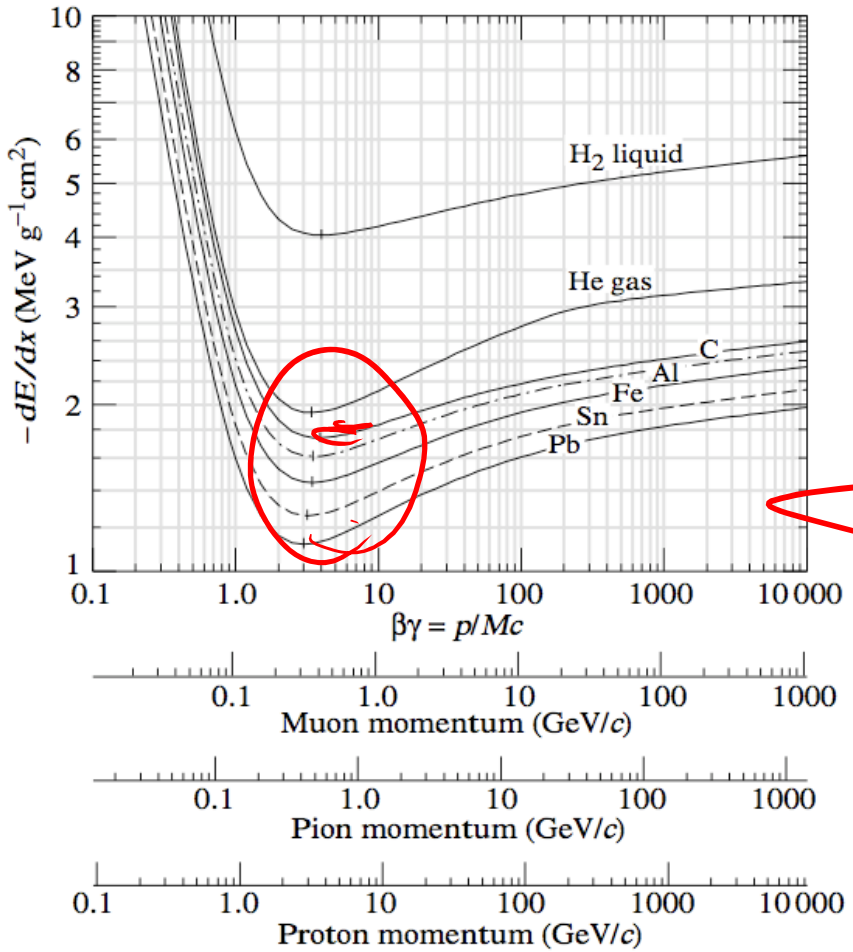
- Atomic electron is knocked free from the atom.
- The remaining atom is now an ion or left in an excited state (will decay by emitting a photon)





Ionization: Bethe-Bloch formula

$$-\frac{dE}{dX} = 4\pi N_A r_e^2 m_e c^2 z^2 \frac{Z}{A}\frac{1}{\beta^2}\left[\frac{1}{2}\ln\frac{2m_e c^2\beta^2\gamma^2 T_{\max}}{I^2} - \beta^2 - \frac{\delta}{2}\right]$$



- where β, γ, relate to particle speed, z is the particle's charge..
- The other factors describe the medium (Z/A, I, δ), or are physical constants.
- dE/dx units is MeV cm²/g
- ΔE= dE/dx * Δx * ρ
- A value to keep in mind is 2 MeV.g⁻¹.cm²

Use this for HW problem

Estimating stopping distance easy way

$$-\frac{dE}{dX} = 4\pi N_A r_e^2 m_e c^2 z^2 \frac{Z}{A}\frac{1}{\beta^2}\left[\frac{1}{2}\ln\frac{2m_e c^2\beta^2\gamma^2 T_{\max}}{I^2} - \beta^2 - \frac{\delta}{2}\right]$$

- MIP value A value to keep in mind is 2 MeV $.g^{-1}.cm^2$

  - Estimate the range of 10 GeV muon in pb
  - Density of pb = 11 g/cm^3

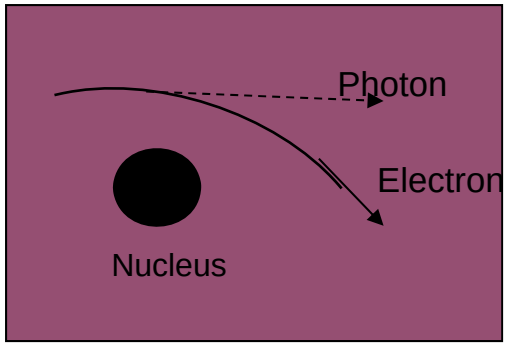
$\frac{10\,GeV}{0.022\,GeV}$ $\frac{dE}{dx} = 2\,MeV\,g^{-1}cm^2\,(\rho)$

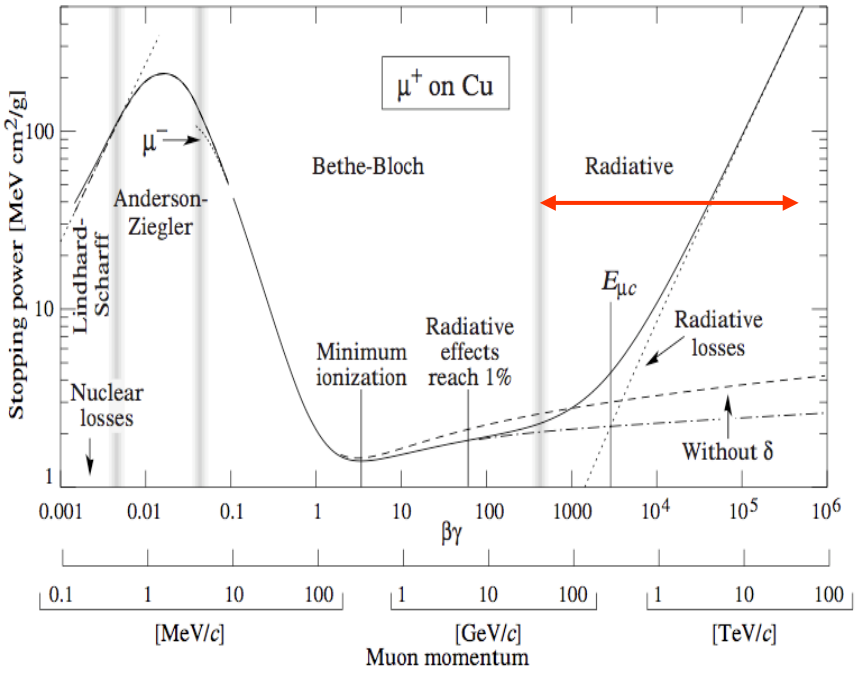
22 MeV/cm.

Charged Particles: Bremsstrahlung

- Radiation of real photons in the Coulomb field of a nuclei of the absorber.

$$-\frac{dE}{dX} = 4\alpha \frac{\rho N_A}{A} Z(Z+1) r_e^2 \ln(183 Z^{-1/3}) E$$





- Define X_0 the radiation length: length during which the particle looses a fraction *e* of its initial energy

$$-\frac{dE}{E} = \frac{dX}{X_0}$$

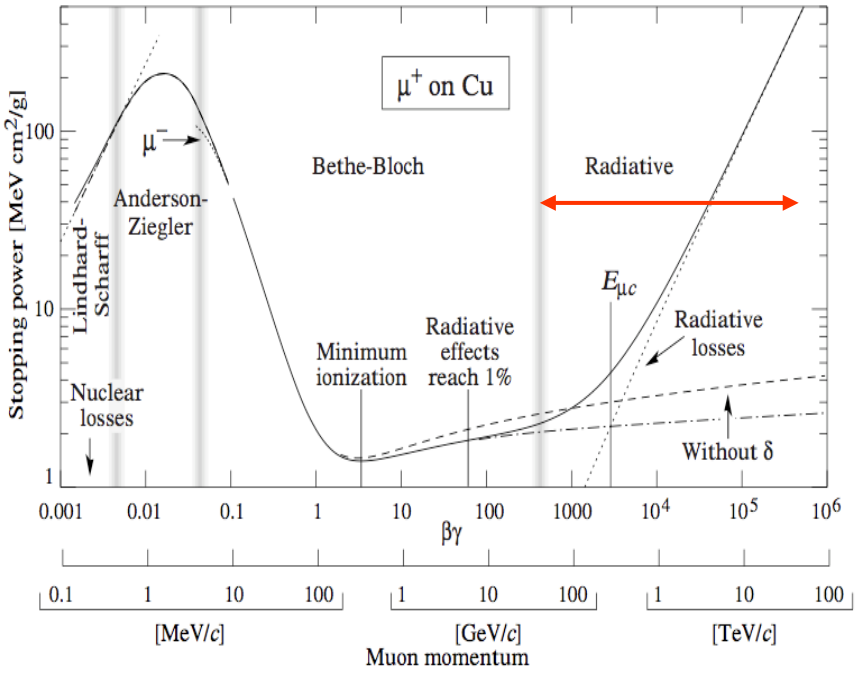
- Critical energy ($E_{\mu c}$) is the energy at which Bremstrahlung loss equal ionization loss

$E_{\mu c}$(e- Cu)=20 MeV

$E_{\mu c}$(μ- Cu)=800 GeV

Exercise (skipped as Exercise but discussed)

- Mostly we are just dealing w/ electrons even for photons
- Radiation length in EM cal
- ~steel/W (tungsten) ~2 cm



- Define X_0 the radiation length: length during which the particle looses a fraction e of its initial energy

$$-\frac{dE}{E} = \frac{dX}{X_0}$$

Estimate energy "leakage" (non released energy)

if detector thickness is 30 cm

LEADS TO EXPONENTIAL prob of survival w/ depth x

$$N \propto e^{-X/X_0}$$

Charged particles: Multiple Coulomb Scattering
(both for ionization and radiation)

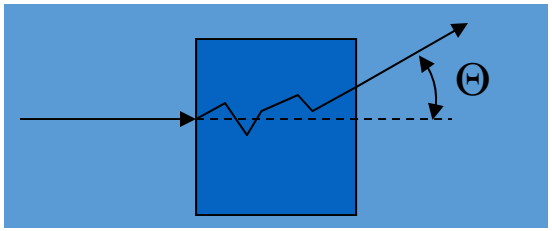
Detectors scatter particles even without much energy loss…

- MCS theory is a statistical description of the scattering angle arising from many small interactions with atomic electrons.
- MCS alters the direction of the particle in average.
- Most important at low energy.

$$\langle \Theta \rangle = 0$$

$$\sigma_\Theta = \frac{13.6\ \ MeV}{\beta cp}\, z\sqrt{x/X_0}\left[1+0.038\ \ \ln\left(x/X_0\right)\right]$$

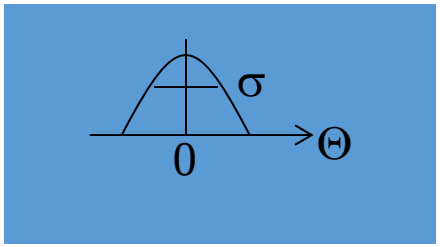
details not important



β is particle speed,

z is its charge,

X_0 is the material's Radiation Length.



Qualitative effect: smears momentum direction!!! as random walk

Charged particles- Cerenkov radiation

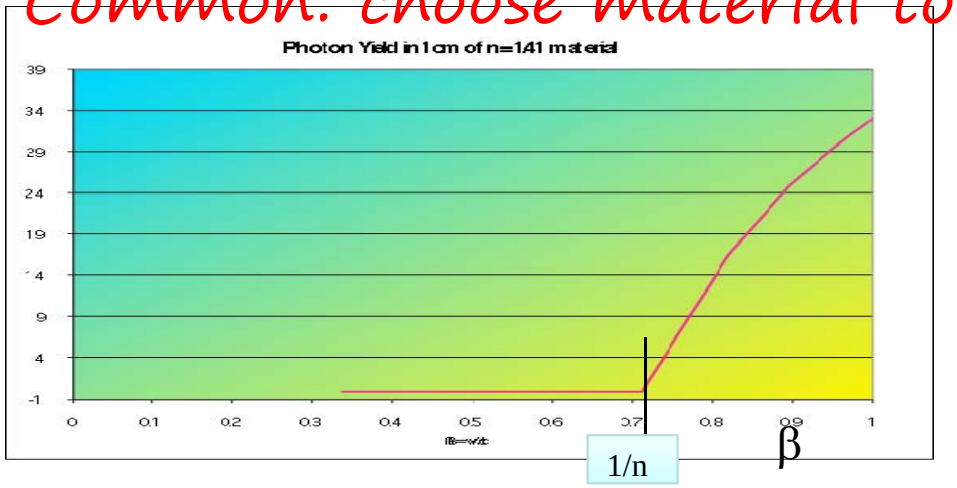
ANOTHER RELATIVISTIC EFFECT!

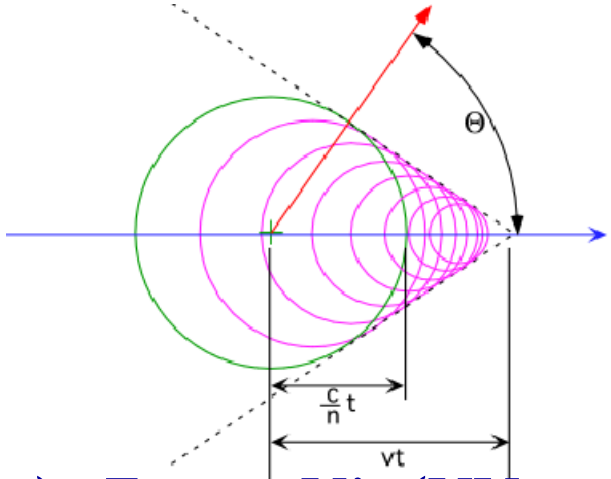
The electric field of a particle has a long-rang interaction with material as it passes through continuous medium.

It does create a shock wave that causes the material t emit light when it's speed is large enough

$v = \beta c > c/n$ (index of refraction)

Common: choose material to ID based on mass threshold



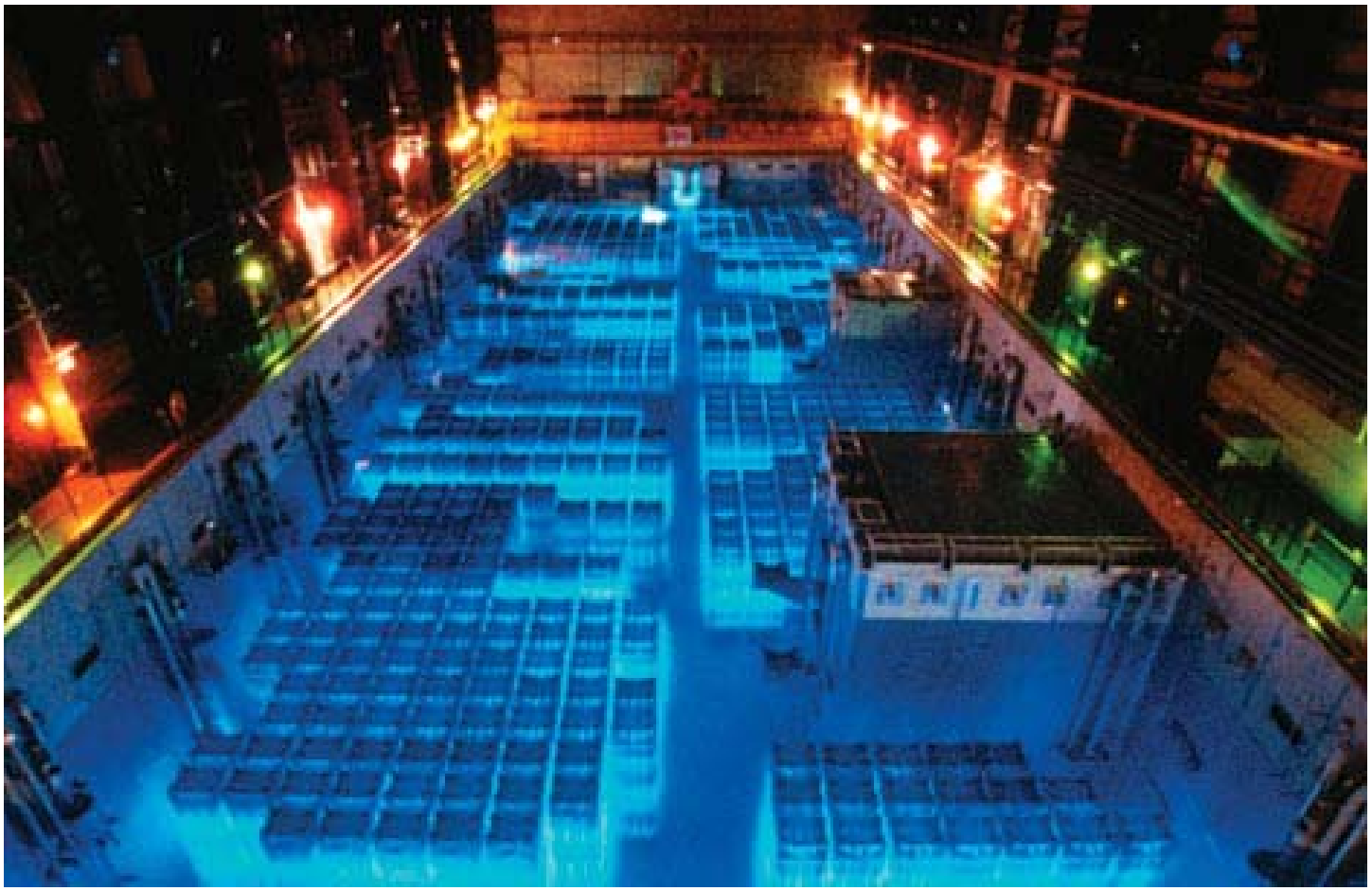


Light emitted at the angle $\Theta = \cos^{-1}(1/\beta n)$. E_{light}~ eV's (UV to visible)

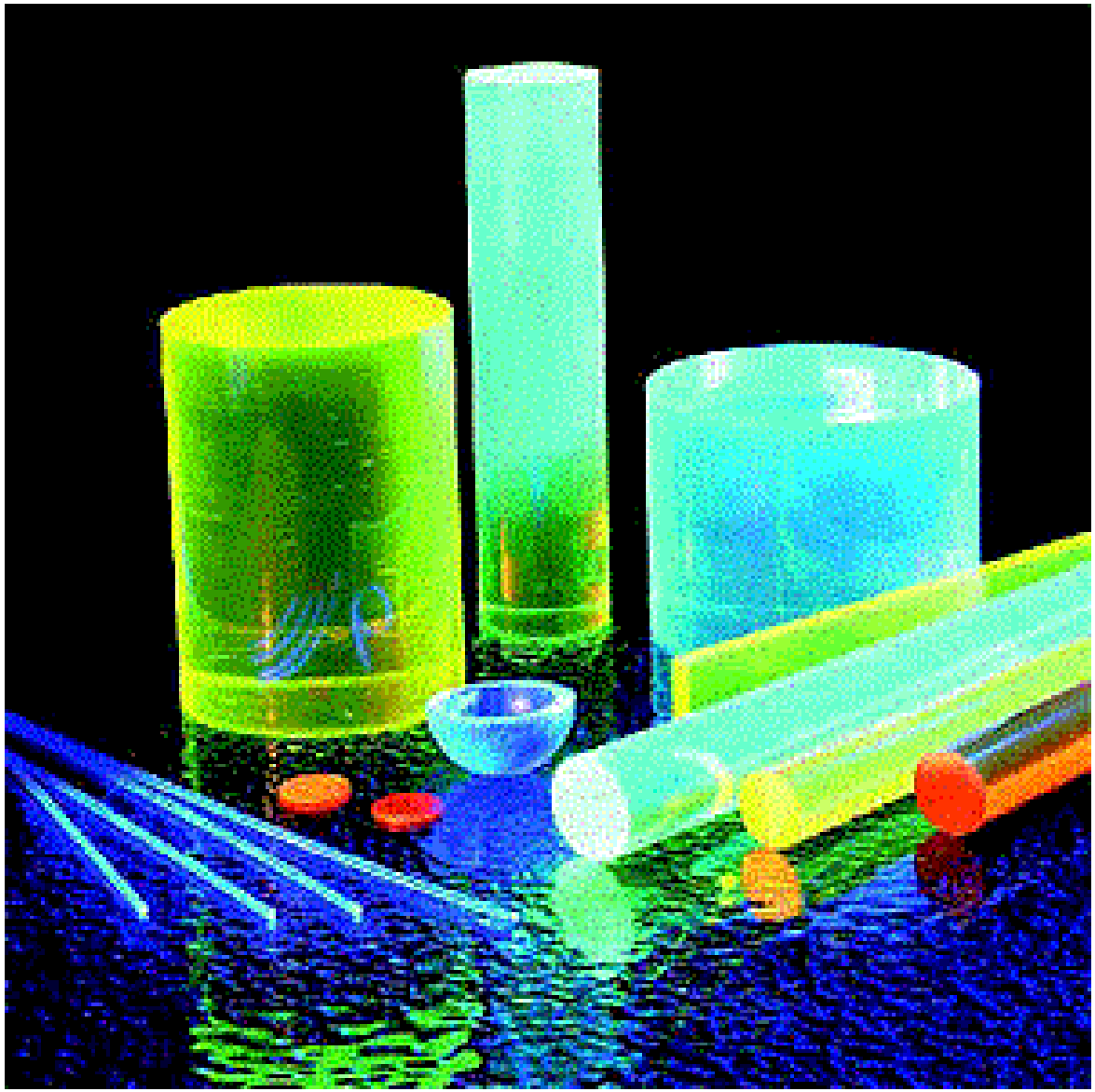
Some calorimeters use this in place of scinttilators

-(similar to scintillation in "form" usually transparent materials)

Cerenkov light produced by fuel elements of a nuclear power plant.



Solids





Liquid

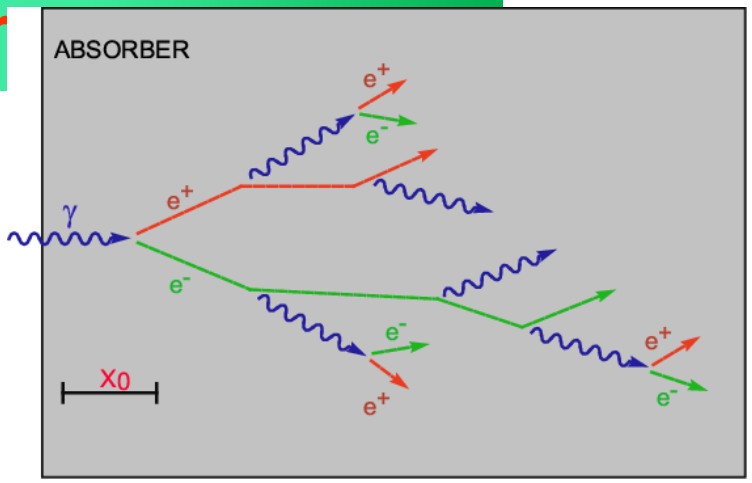
Saint Gobin Inc

Particle Detectors: Calorimeter

Used to measure energy.

Long matter EM alternating Brems. &

Pair production → "Shower"



Suppose an initial photon of energy E_o

After t radiation lengths, the average energy of secondary is: $E(t) = E_0 / 2^t$

The shower stops when $E(t_{\max}) = E_c$ $\quad t_{\max} = \dfrac{\ln(E_0/E_c)}{\ln 2} \sim 10X_0$

The number of secondaries is $N_{\max} = \exp(t_{\max} \ln 2) = E_0 / E_c$

The energy resolution of the calorimeter therefore goes as $\dfrac{\Delta E_0}{E_0} = \dfrac{1}{\sqrt{E_0}}$

Energy RESOLUTION Dependent on number of shower particles

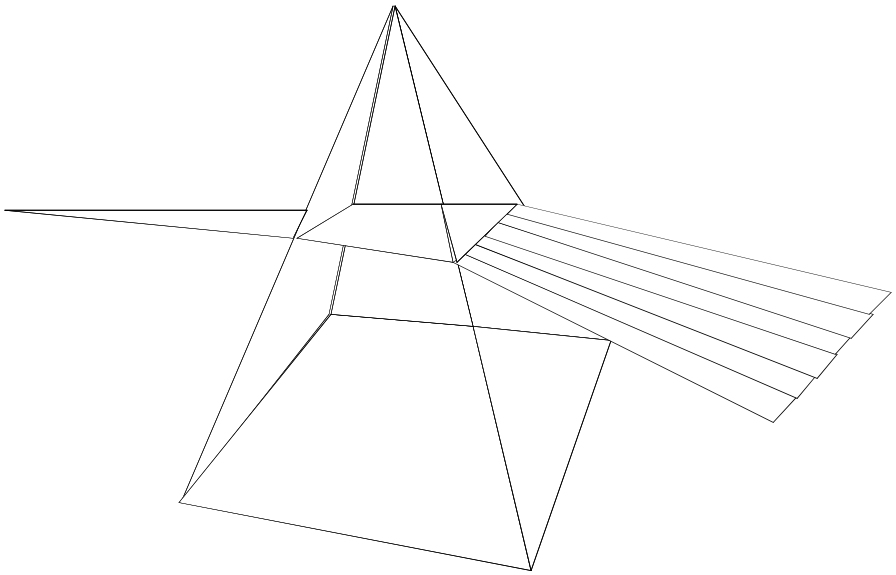
Poisson process so reso RMS (ΔE or σ_E) depends on $\sqrt{N}$ → $\sqrt{E}$

TRACKING DETECTORS magnetic "spectrometers"

TRACK TRAJECTORIES → BEND
BETTER RESO WHERE BEND IS LARGER→ low MOMENTUM

TYPICALLY Ch tracking detectors placed in Magnet

Just as light of different wavelength is bent differently by a prism...



Nature lets us measure the Momentum of a charged particle by seeing how much its path is deflected by a magnet.

$$p(GeV) = 0.3\, B(T)\rho(m)$$

